Meridian

Assembly and Logistics Centre

Coordinated concept and design development

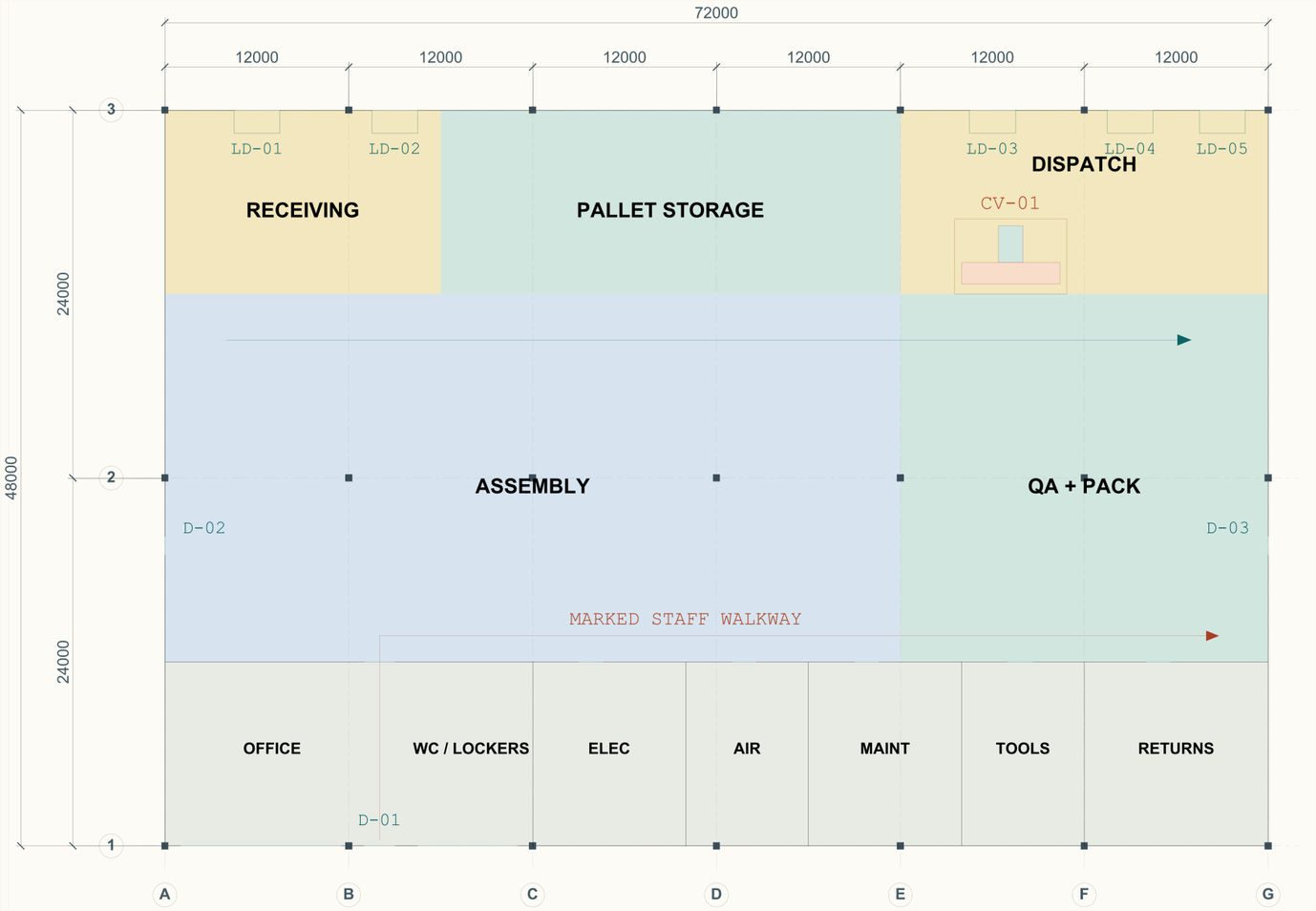
This portfolio study traces materials from receiving through assembly, quality checks and dispatch.

The drawings coordinate the hall, loading docks, steelwork, equipment and building services.

3,456 m²
HALL FOOTPRINT

5
LOADING DOCKS

37
DRAWING SHEETS



G000 / Project master excerpt / Refer to sheet for scale

01 REV 01 COORDINATED CONCEPT

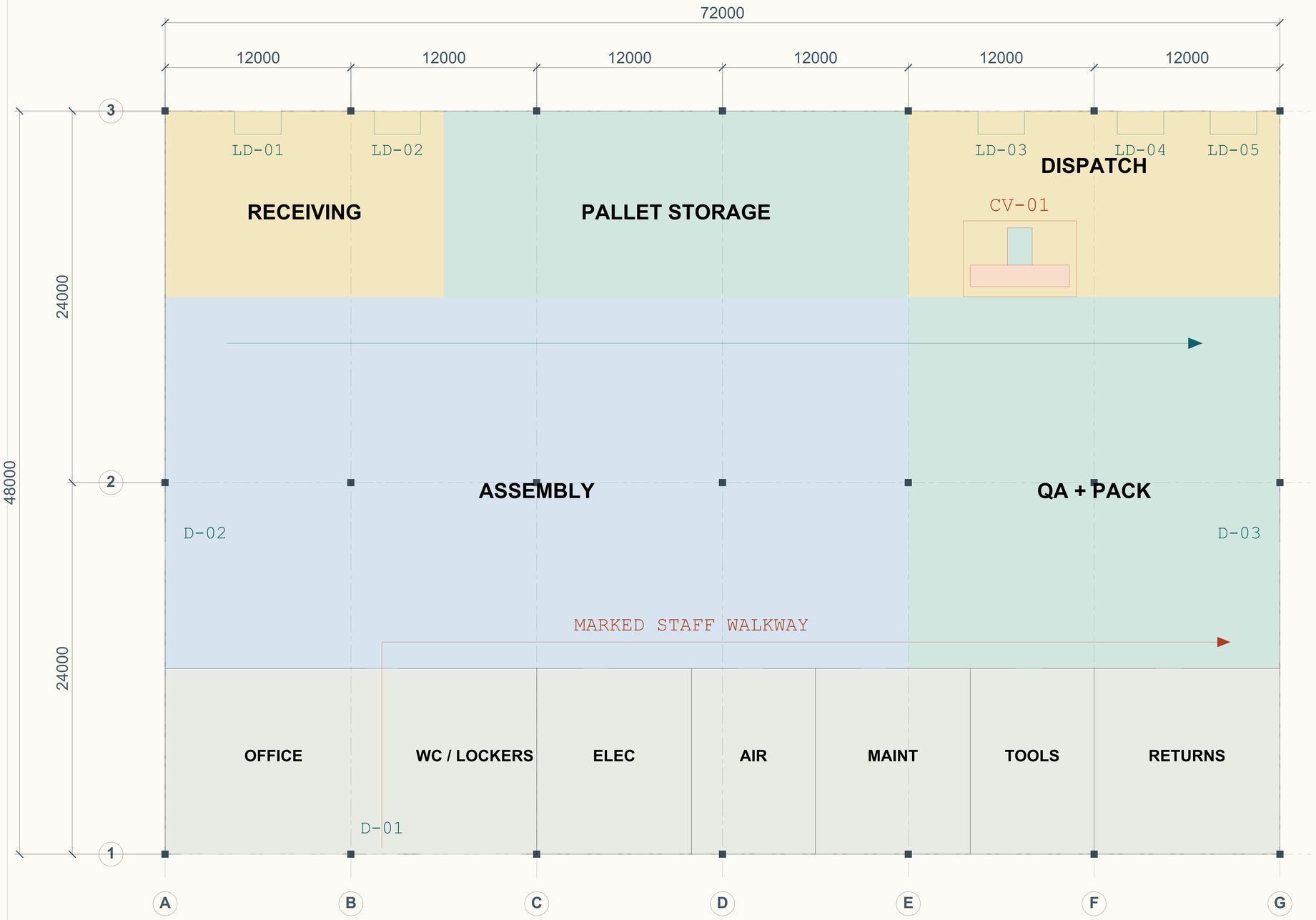
26 sheets including the A0 project master

02 REV 02 DESIGN DEVELOPMENT

11 A3 sheets covering interfaces and study checks

Integrated project master

SITE 14,520 M² / HALL 3,456 M² / CROSS-DISCIPLINE DRAWING REGISTER



ISSUE REGISTER / 26 SHEETS

SHEET	DRAWING
G000	Integrated project master
G001	Project overview
C101	Site + circulation
A101	Ground-floor plan
P101	Material flow + equipment
S101	Roof + structural grid
A201	Elevations + section
M101	Utilities coordination
E601	Electrical distribution
C102	Civil levels and drainage
A102	Staff and support rooms
A202	End elevations and long section
A301	Loading dock interface
A601	Openings and room finishes
S102	Foundations and slab coordination
S201	Portal frames and stability
S301	Structural interfaces
A501	Building envelope details
M201	Air systems and elevations
P201	Water and sanitary drainage
F101	Egress and fire access
F201	Fire system interfaces
E201	Lighting and low voltage
E602	Distribution and bonding
P102	Vector equipment integration
Q001	Coordination and acceptance

CORE DATUMS

72 × 48 m hall / 12 m frame bays
Ground floor +0.00 / dock apron −1.20
Eaves +8.00 / twin ridges +10.40

DESIGN STAGE

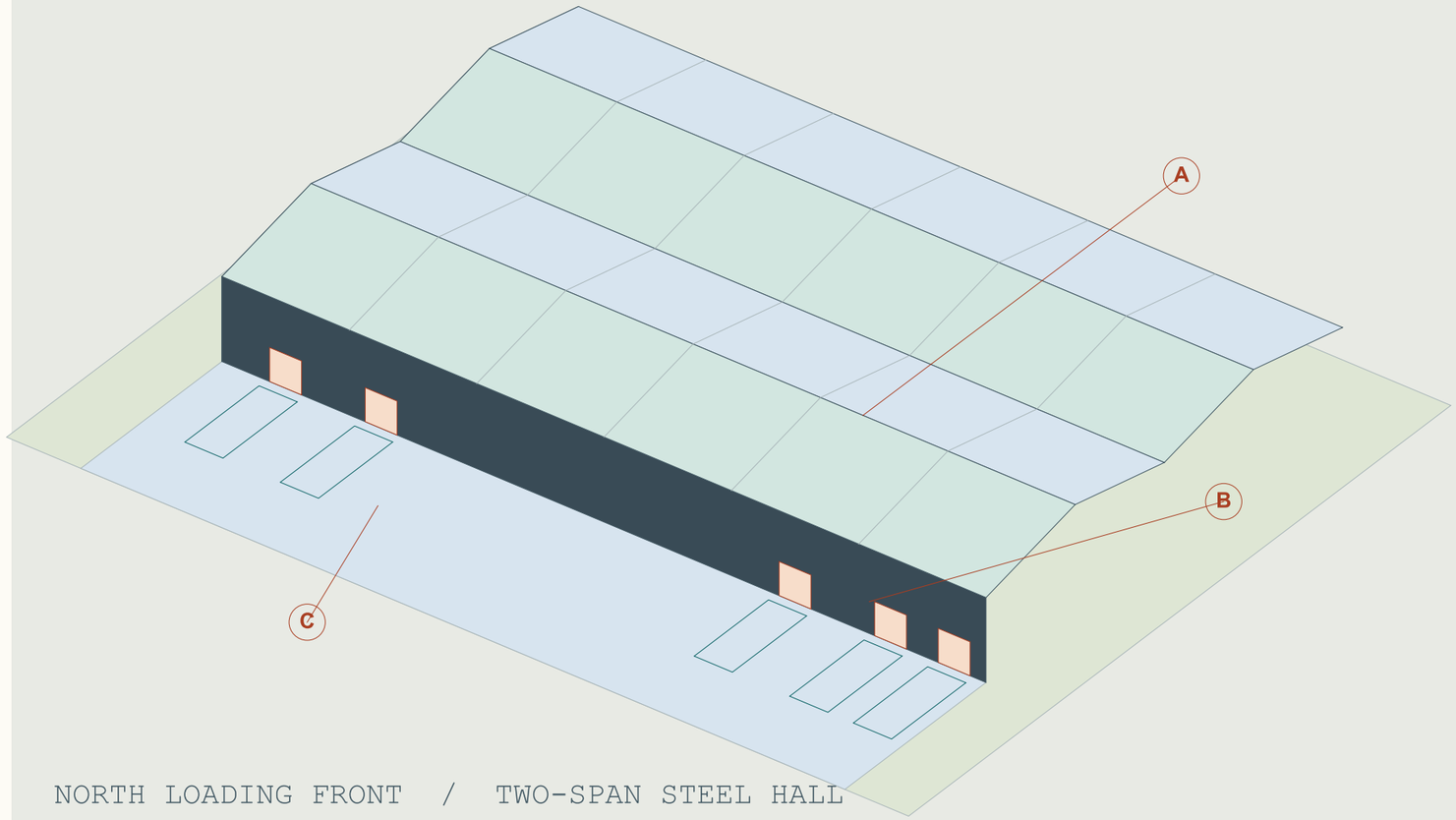
Coordination concept / Q001

Project overview

01 / INDUSTRIAL CAMPUS STUDY 02 / COORDINATED MULTIDISCIPLINARY SET

01 BUILDING AXONOMETRIC

A / TWIN PITCHED SPANS
B / FIVE DOCK POSITIONS
C / NORTH TRUCK COURT



NORTH LOADING FRONT / TWO-SPAN STEEL HALL

PROJECT PARAMETERS

3,456 m²

GROSS FLOOR AREA / 72 × 48 m

SHEET	DRAWING INDEX
G001	Project overview
C101	Site + circulation
A101	Ground-floor plan
P101	Material flow + equipment
S101	Roof + structural grid
A201	Elevations + section
M101	Utilities coordination
E601	Electrical distribution

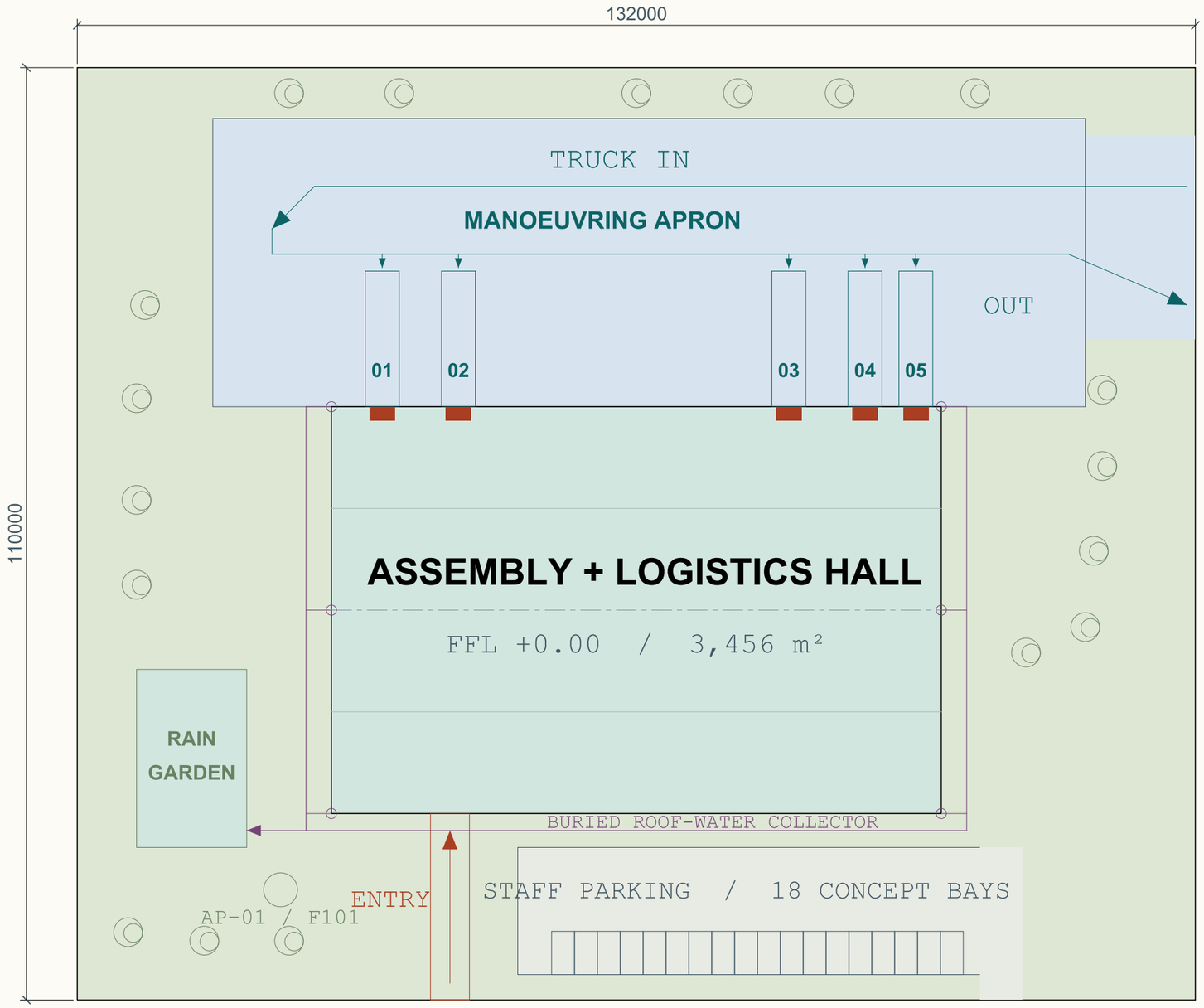
Full 26-sheet drawing register on G000.

PROJECT INTENT

Receiving to dispatch in one hall. Public entry and truck access are on separate sides of the site.

Site + circulation

NORTH TRUCK COURT / STAFF APPROACH FROM SOUTH / LANDSCAPE SETBACKS



SITE DATA / CONCEPT

METRIC	VALUE
Site footprint	14,520 m²
Hall footprint	3,456 m²
Site coverage	23.8%
Truck apron depth	34 m
Dock positions	5
Staff parking	18 bays

01 / TRUCKS

Separate in and out lanes at the east throat. Reverse from the court to docks LD-01 through LD-05.

02 / PEOPLE

Walk from the south gate to D-01. Staff parking uses a separate spur.

03 / WATER

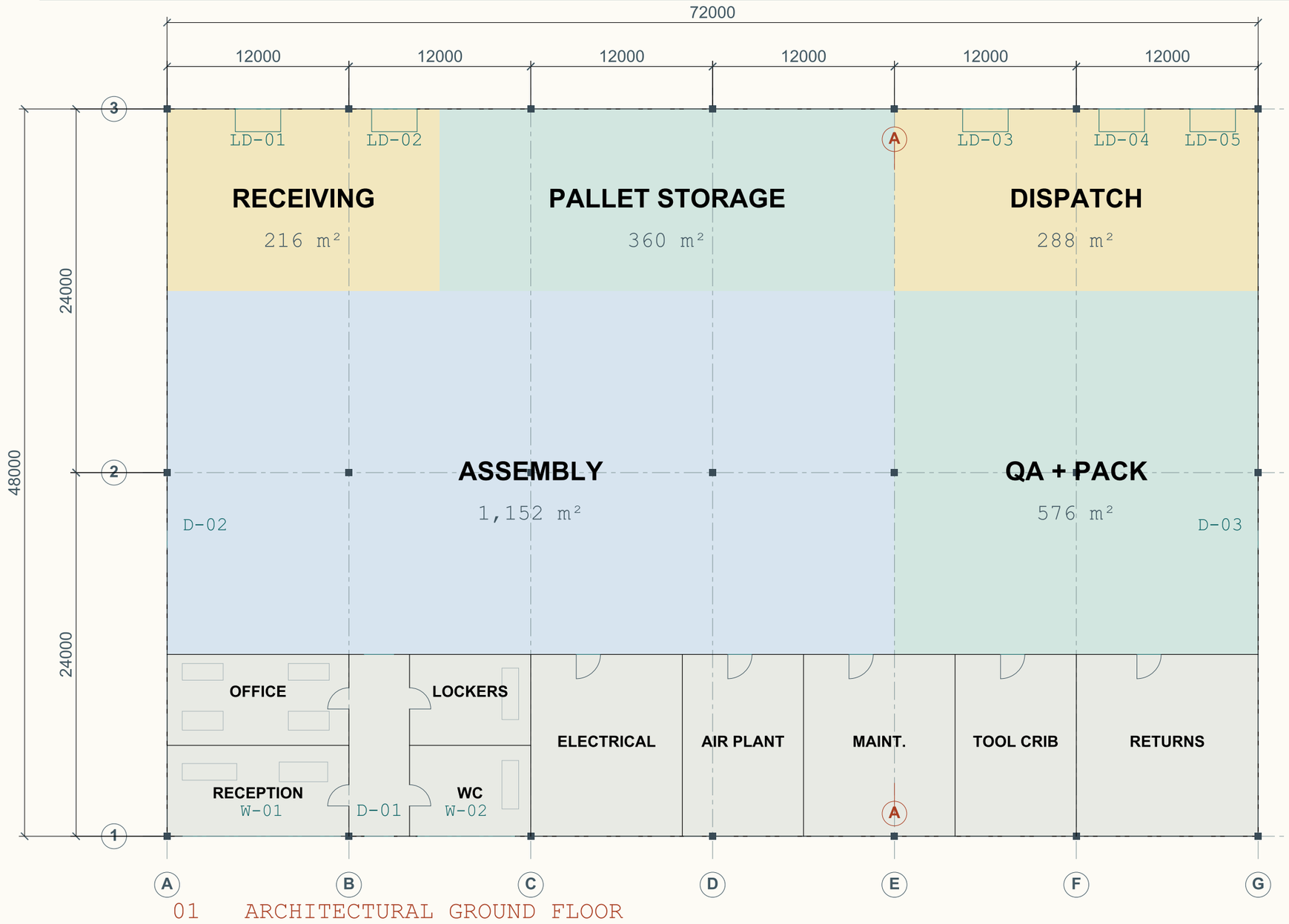
West landscape reserve receives roof runoff in the concept layout.

N

0 10 20 30 m

Ground-floor plan

GROUND DATUM +0.00 / DIMENSIONS IN MM / OPENING TAGS COORDINATED WITH C101 + A201



01 ARCHITECTURAL GROUND FLOOR

PROGRAM AREA

ZONE	M²
Receiving	216
Pallet Storage	360
Dispatch	288
Assembly	1,152
Qa + Pack	576
Office + Staff	288
Support + Services	576
TOTAL	3,456

OPENING REGISTER

MARK	QTY	CLEAR W
LD-01–05	5	3,000 mm
D-01	1	2,000 mm
D-02 / 03	2	1,200 mm
W-01 / 02	2	10.5 / 6.5 m

SECTION A–A

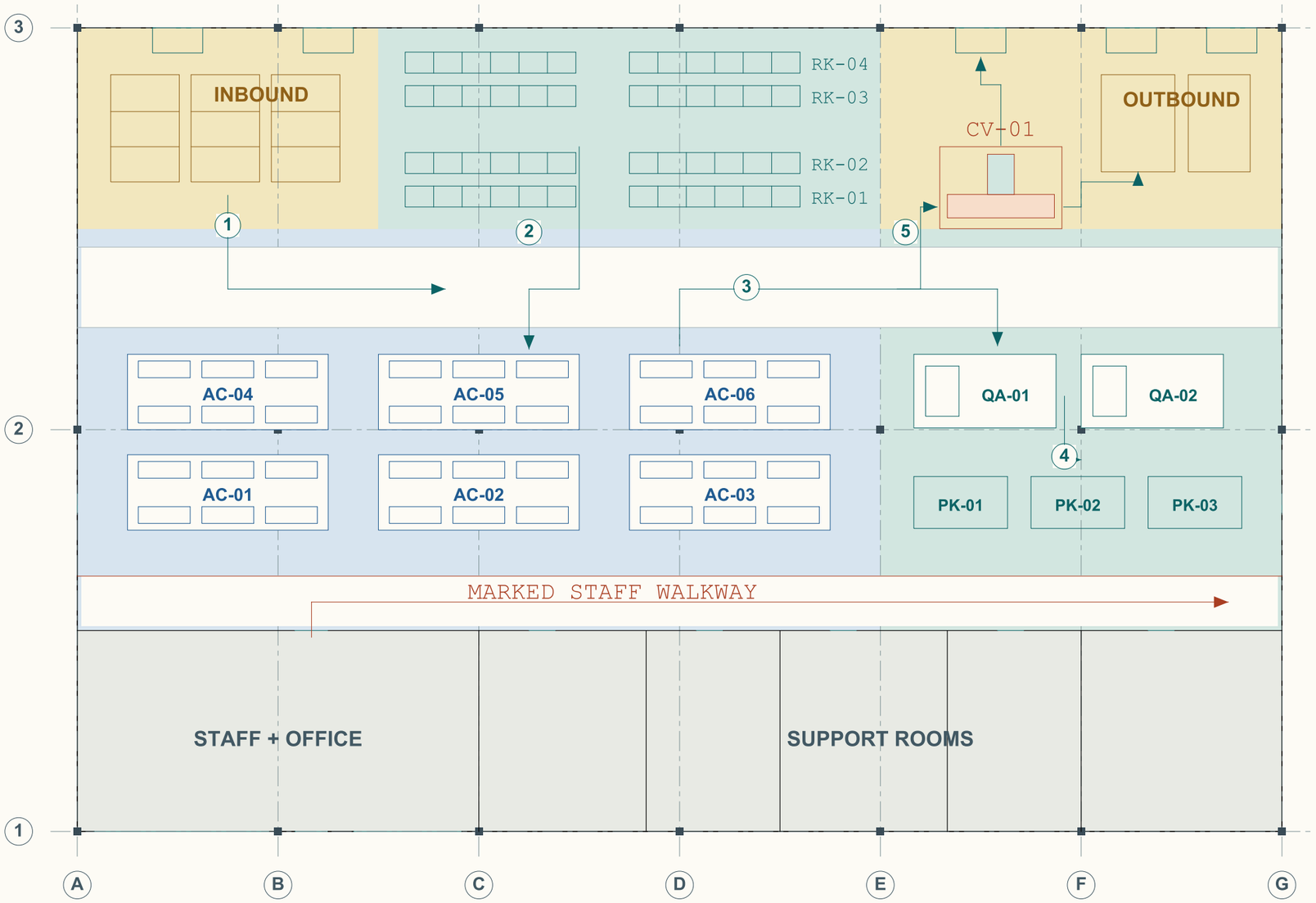
Cross-hall cut at grid E.
Refer to sheet A201.

N

0 5 10 15 m

Material flow + equipment

RECEIVE → STORE → ASSEMBLE → TEST → PACK → DISPATCH



01 EQUIPMENT AND MATERIAL TRANSFER

EQUIPMENT REGISTER

ID	QTY	ASSET
AC-01–06	6	Assembly cells
QA-01–02	2	Test stations
PK-01–03	3	Packing benches
RK-01–04	4	Rack rows
LD-01–05	5	Loading docks
CV-01	1	Vector transfer

MATERIAL SEQUENCE

1 Receive + stage
2 Store + replenish
3 Assemble + test
4 Pack to outbound lanes
5 Transfer + release to dispatch

MOVEMENT KEY

Teal / material transfer
Rust / staff route
Central aisle band / 4.8 m

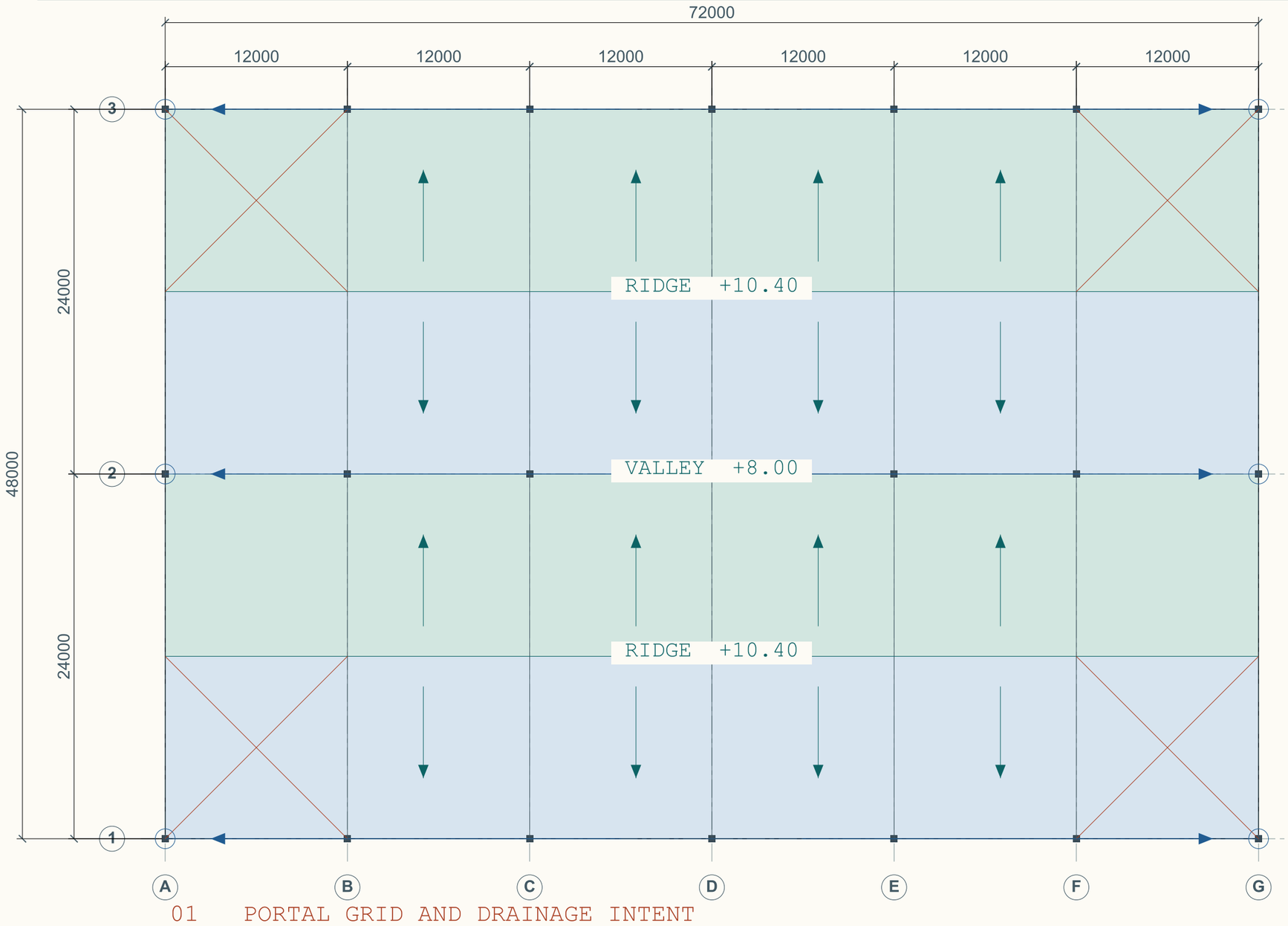
SYSTEM REFERENCES

Vector equipment interface: P102
Utility drops: M101 / E602

0 5 10 15 m

Roof + structural grid

7 PORTAL FRAMES / 6 LONGITUDINAL BAYS / EAVES, RIDGES AND VALLEY



01 PORTAL GRID AND DRAINAGE INTENT

PRIMARY GEOMETRY

ITEM	VALUE
Transverse spans	2 × 24 m
Longitudinal bays	6 × 12 m
Portal lines	7
Column points	21
Eave / valley	+8.00 m
Roof ridges	+10.40 m
Nominal roof fall	1:5

ROOF BRACING

Rust crosses mark roof-plane bracing zones at the end bays.

WATER COLLECTION

Two eaves plus central valley.
Six outlet locations shown.
Roof runoff route on C101.

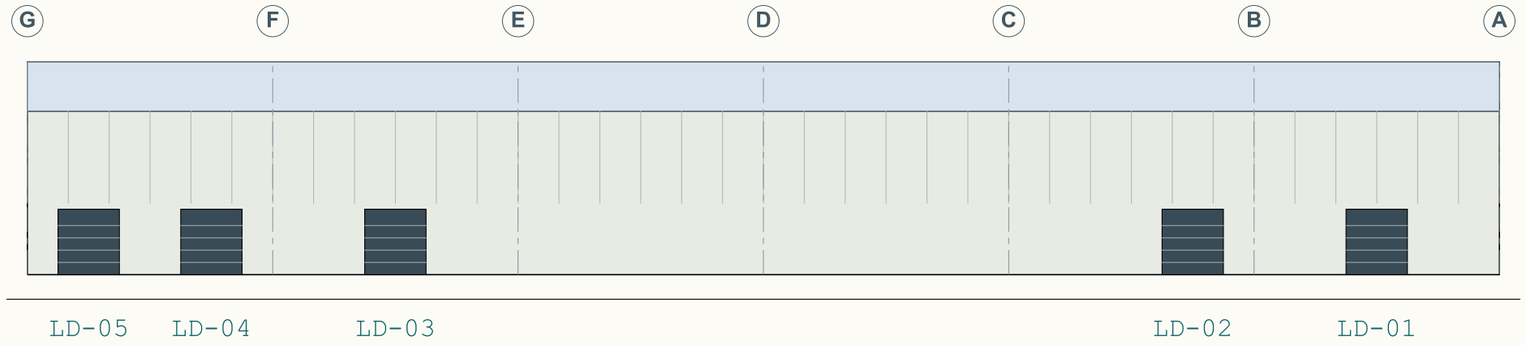
FRAME SECTION

Refer to section A–A on A201.

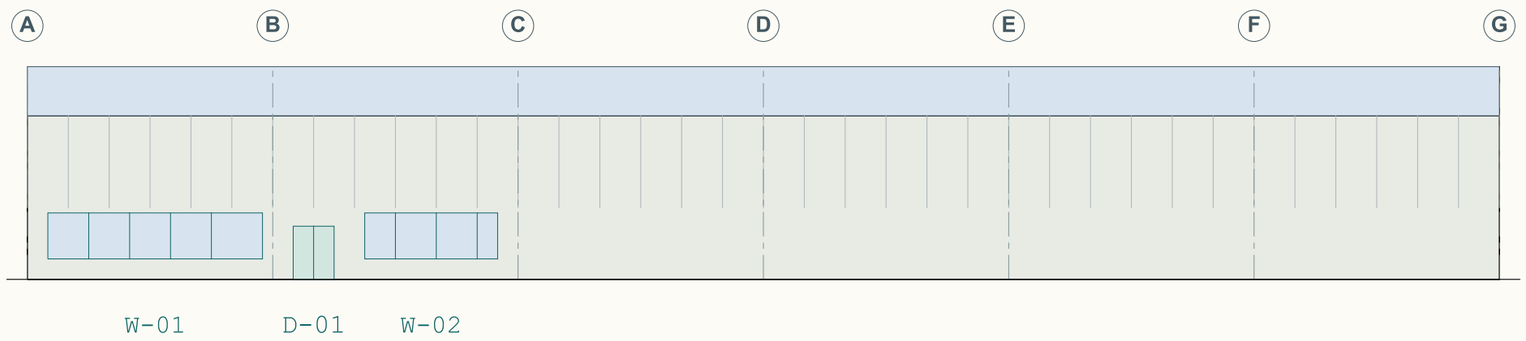
0 5 10 15 m

Elevations + section

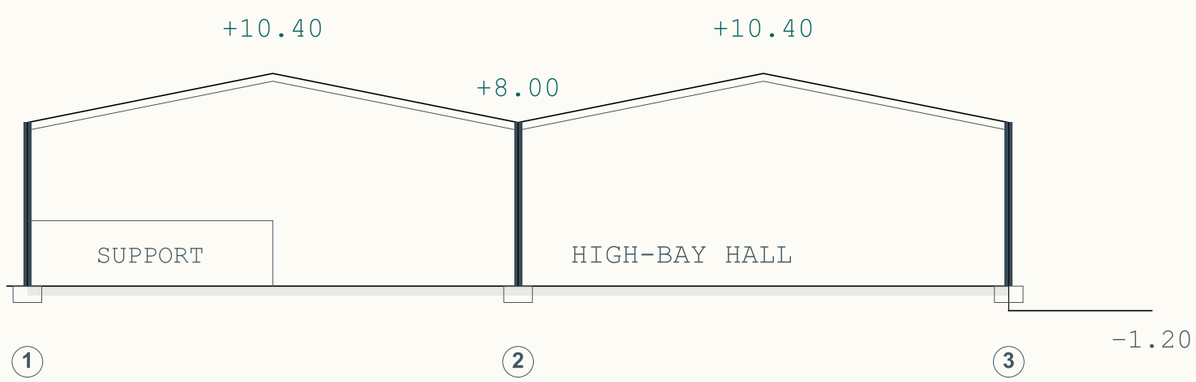
FACADE OPENINGS / SHARED LEVELS / CROSS-HALL SECTION AT GRID E



01 NORTH / LOADING ELEVATION



02 SOUTH / STAFF ENTRY ELEVATION



03 SECTION A-A / GRID E

01 / DOCK FACADE

Five 3.00 m clear openings. Dock floor is 1.20 m above the conceptual truck apron.

02 / STAFF FRONT

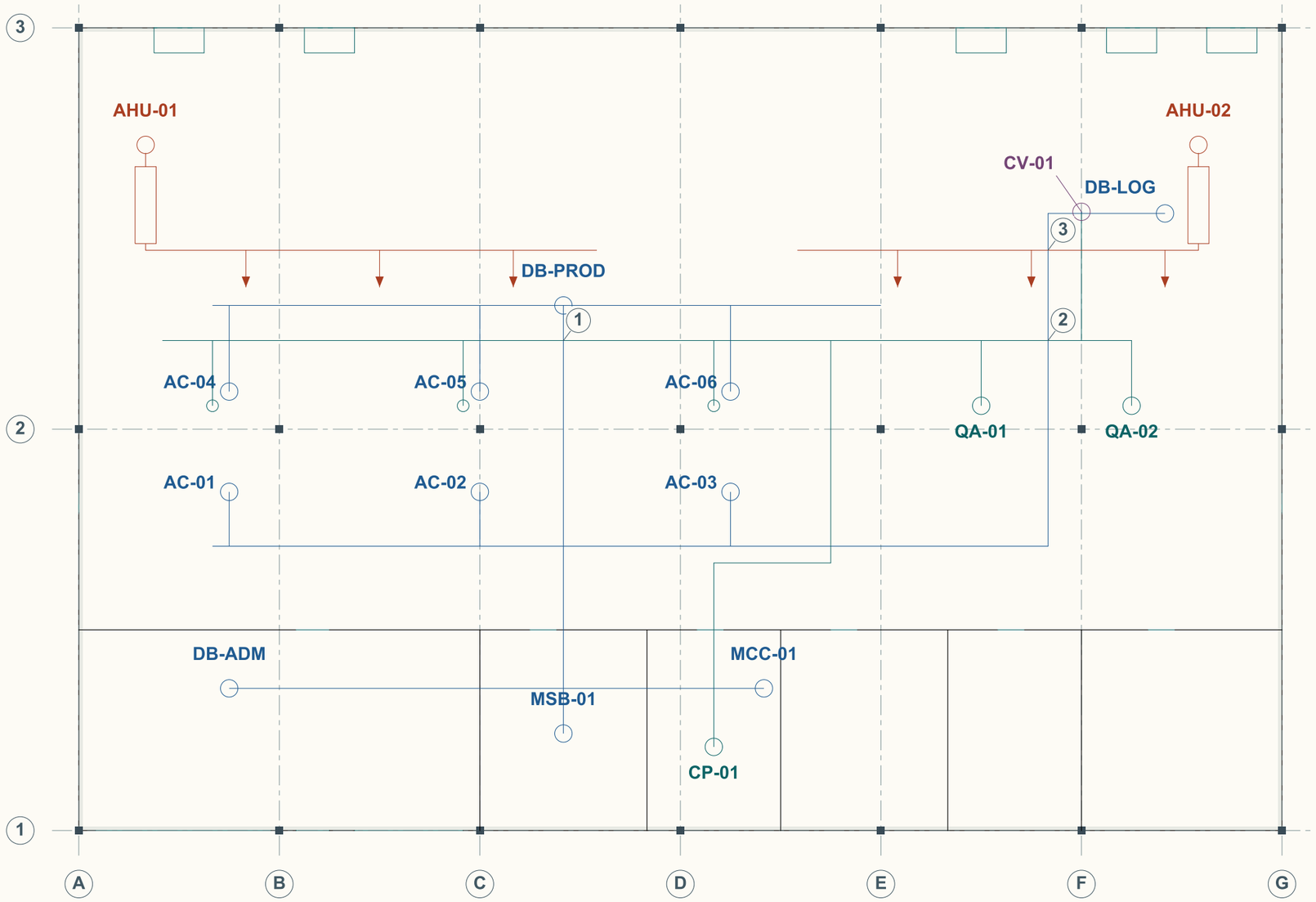
Public entrance D-01 is aligned with the south pedestrian route. Glazing serves the office spine.

03 / LEVEL SCHEDULE

DATUM	LEVEL (M)
Ridge	+10.40
Eave / valley	+8.00
Ground floor	+0.00
Dock apron	−1.20

Utilities coordination

SERVICE ROUTES BY SYSTEM / EQUIPMENT DROPS / ELEVATION BANDS AT SHARED CROSSINGS



01 OVERHEAD UTILITIES COORDINATION

ROUTING LEGEND

- Electrical tray
- Compressed air
- Ventilation duct

TARGET ROUTE LEVELS

SYSTEM	CENTRELINE
Ventilation	+6.90 m
Cable tray	+6.20 m
Air header	+5.30 m

CROSSING REGISTER

1 Tray / air at production spur
2 Tray / air at dock riser route
3 Tray / duct at east supply

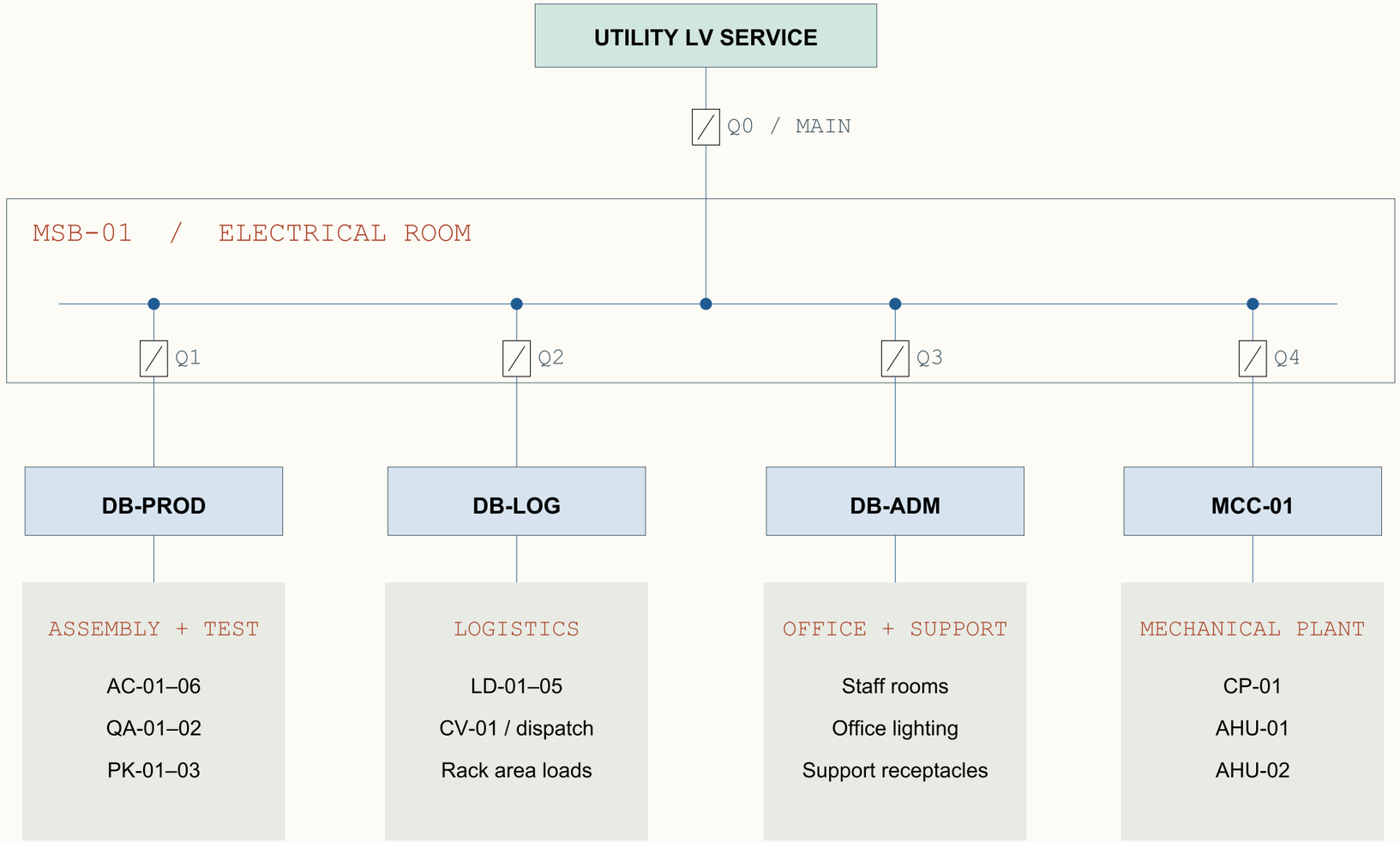
INTERFACES

Plan levels are nominal targets.
Match board tags to E601 / E602.
CV-01 power + air at (60,37).

0 5 10 15 m

Electrical distribution

RADIAL TOPOLOGY / MAIN SWITCHBOARD AND FOUR DISTRIBUTION BRANCHES



01 MAIN DISTRIBUTION / TYPICAL BRANCH HIERARCHY

Feeder labels are functional references for the coordination plan.

PANEL ALLOCATION

TAG	AREA
MSB-01	Electrical room
DB-PROD	Production zone
DB-LOG	Logistics zone
DB-ADM	Office / support
MCC-01	Air plant

DEVICE KEY

Q0 / Main protective device
Q1–Q4 / Feeder protection
Blue line / Power path

DESIGN BASIS

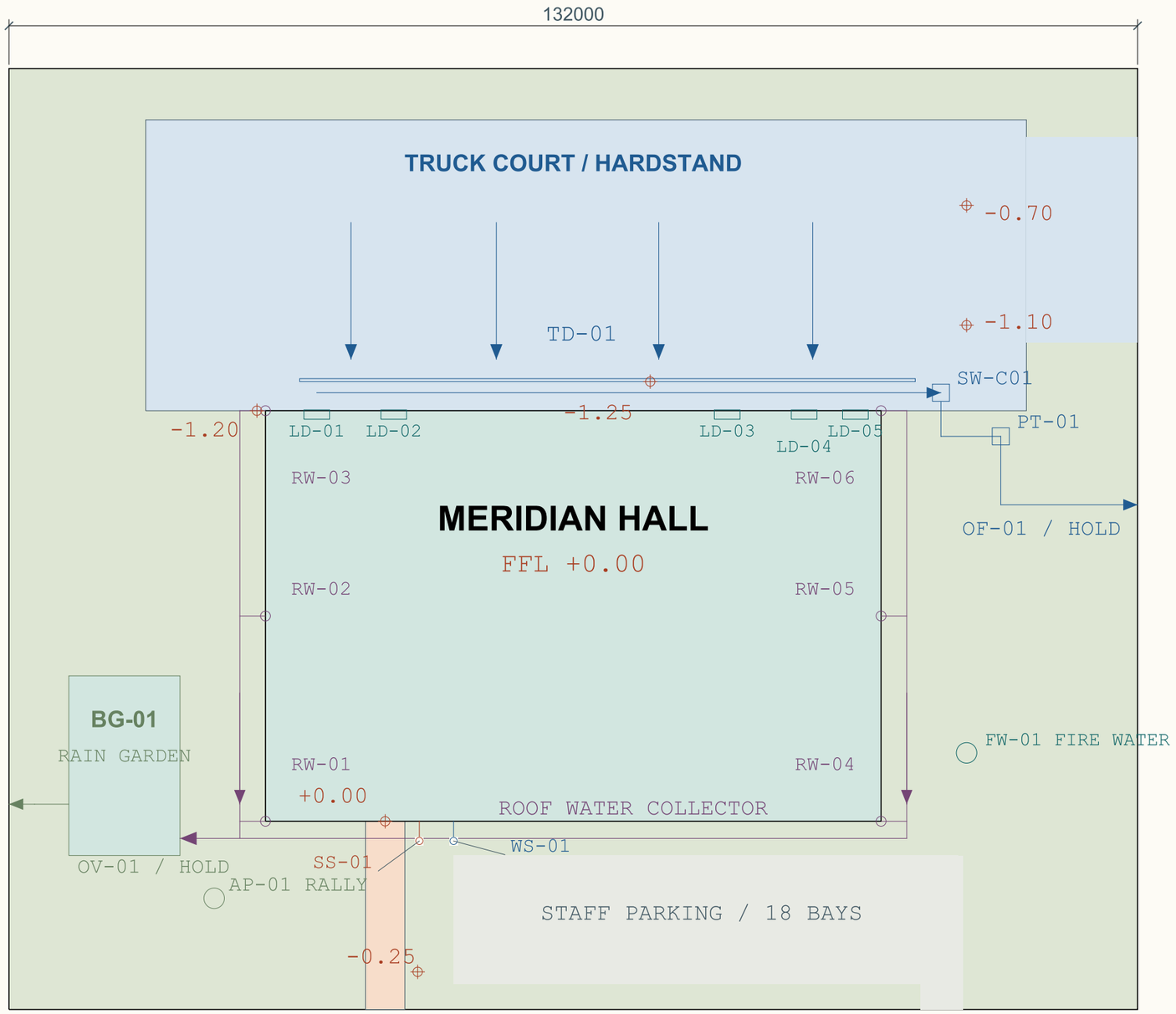
Concept distribution only.
Confirm supply, loads, ratings and coordination during design.

PLAN INTERFACE

Physical routes M101. Branches E602.

Civil levels and drainage

CONCEPT LEVELS / ROOF WATER AND TRUCK-YARD RUNOFF ON SEPARATE ROUTES



LEVEL SCHEDULE / CONCEPT

REFERENCE	M
Hall finished floor	+0.00
South approach control	-0.25
Dock apron at building	-1.20
Yard capture surface	-1.25
East gate approach	-0.70

01 / ROOF WATER

Six roof leaders to BG-01.
Provide overflow OV-01.
Verify infiltration and discharge.

02 / TRUCK YARD

TD-01 to SW-C01 and PT-01.
Treatment and outfall OF-01 are
held for hydraulic design and approval.

03 / PAVEMENT + UTILITIES

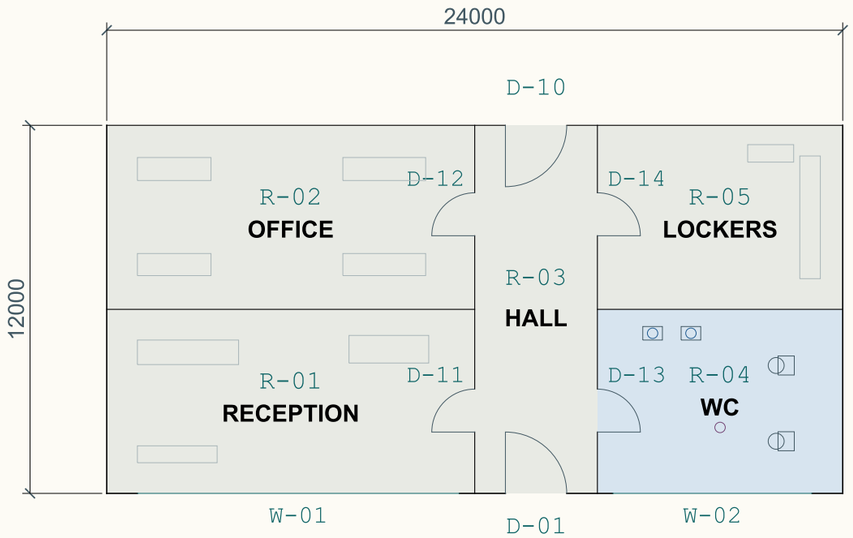
Heavy duty court / light duty parking.
Confirm utility crossing, pavement,
invert levels and vehicle sweep.

N

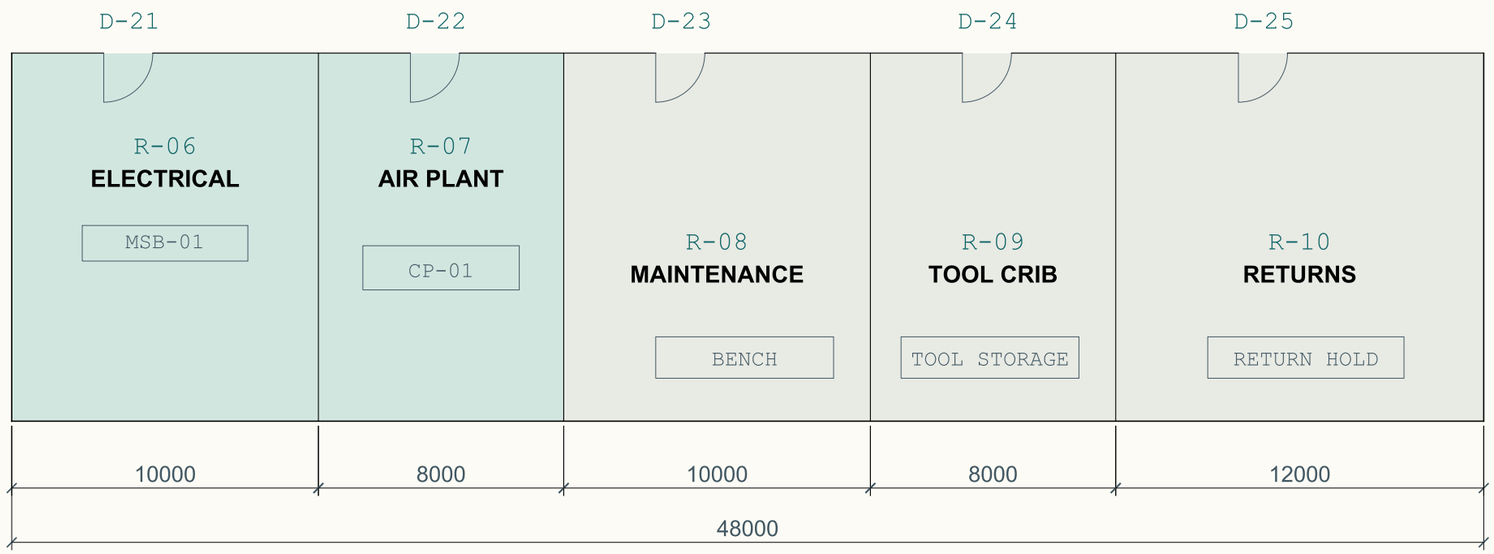
0 10 20 30 m

Staff and support rooms

ENLARGED SOUTH SPINE / ROOM TAGS AND DOOR SWINGS / REFER TO A601



01 STAFF + RECEPTION SUITE



02 ENGINEERING + SUPPORT SUITE / WEST EDGE AT GRID C

STAFF SUITE / 288 m² GROSS

D-01 aligns with the south walk.
D-10 links the hall to production.
Protect the door clear zones.

COORDINATION

WC fixtures on P201.
Exterior openings on A201/A601.

SUPPORT SUITE

576 m² gross along the south wall.
Floor datum +0.00.
Service plant locations per M101.

ROOM SCHEDULE / A601

Interior marks D-11 to D-14 and D-21 to D-25 are repeated in the opening register.

EQUIPMENT INTERFACES

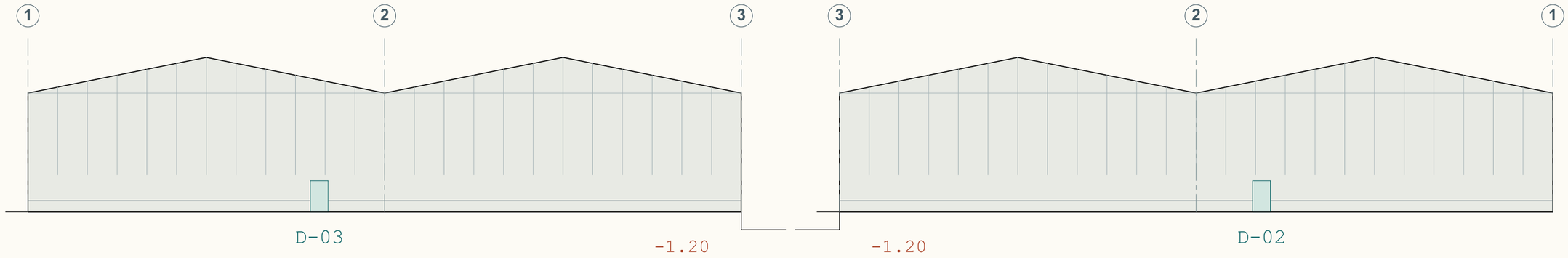
MSB-01 / electrical supply.
CP-01 / compressed air.
Envelope zones are schematic.

RELEASE HOLD

Confirm wall assemblies, door ratings, hardware, fixture access and maintenance clearances.

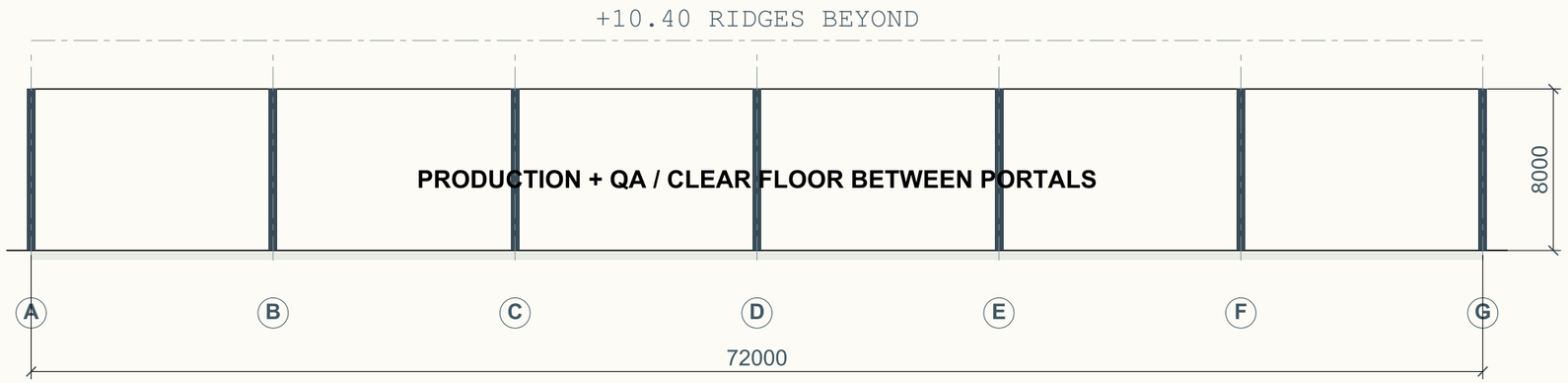
End elevations and long section

SIDE EXIT OPENINGS / FULL RIDGE PROFILES / SECTION B-B ALONG GRID 2



01 EAST / GRID G

02 WEST / GRID A



03 SECTION B-B / LOOKING NORTH ALONG GRID 2

LEVEL COORDINATION

+0.00 finished floor.
+8.00 eaves and central valley.
+10.40 twin roof ridges.

OPENINGS

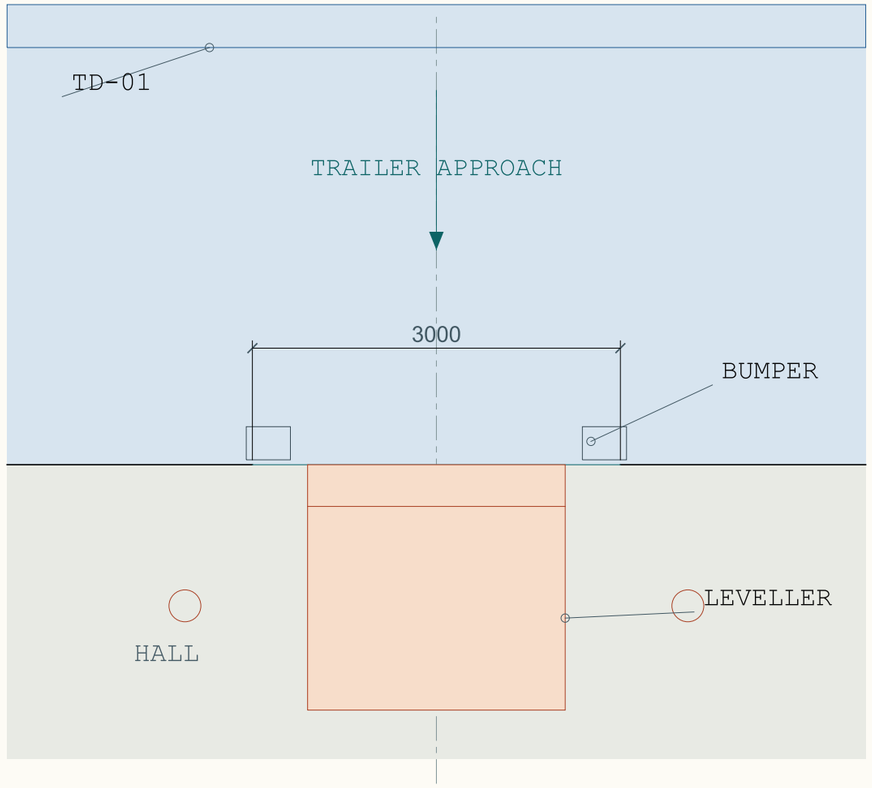
D-02 west / D-03 east.
Common centre Y=19.60 m.
Refer to A101 and A601.

STRUCTURE

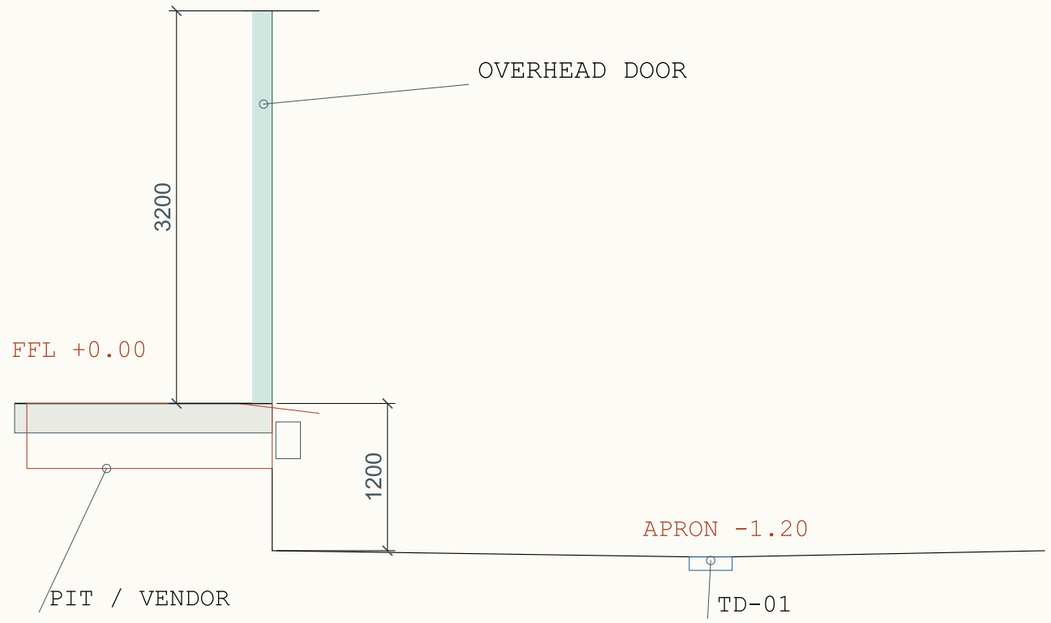
Portal spacing 12 m.
Bracing and frame on S201.

Loading dock interface

TYPICAL NORTH LOADING BAY / LEVELLER PIT AND APPROACH / VENDOR SET-OUT HOLD



01 TYPICAL DOCK PLAN



02 TYPICAL SECTION / THROUGH DOOR CENTRE

DOCK REGISTER / GRID ORIGIN A

BAY	X (M)	USE
LD-01	6.0	Receiving
LD-02	15.0	Receiving
LD-03	54.0	Dispatch
LD-04	63.0	Dispatch
LD-05	69.0	Dispatch

FIXED INTERFACES

3,000 clear width / 3,200 clear height.
1,200 level difference to the apron.
TD-01 drains to the yard system on C102.

PROCUREMENT HOLD

Confirm pit, restraints, seals and bumpers.
Verify actual trailer heights and sweep.
Coordinate final edge/slab with S102.

Openings and room finishes

EXTERNAL AND INTERNAL OPENINGS / SOUTH SPINE ROOM FINISH REGISTER

01 OPENING SCHEDULE / CLEAR NOMINAL DIMENSIONS

ID	QTY	W × H (MM)	LOCATION
D-01	1	2,000 × 2,600	South staff entry
D-02 / 03	2	1,200 × 2,100*	West / east exit
D-10	1	2,000 × 2,100*	Hall to production
D-11 / 12	2	1,400 × 2,100*	Reception / office
D-13 / 14	2	1,400 × 2,100*	WC / lockers
D-21 / 22	2	1,600 × 2,100*	Electrical / air plant
D-23 / 24 / 25	3	1,600 × 2,100*	Maintenance / tool / returns
W-01	1	10,500 × 2,250	South reception
W-02	1	6,500 × 2,250	South staff suite
LD-01 / 02	2	3,000 × 3,200	North receiving
LD-03 / 04 / 05	3	3,000 × 3,200	North dispatch

OPENING NOTES

* Height allowance to confirm in detailed design.
W-01/W-02 sill +1.00 m per A201.
Confirm frame, threshold, hardware, security and required ratings.
LD sets include equipment and drainage interfaces on A301.

02 ROOM + FINISH SCHEDULE

ROOM	USE	M²*	FLR	WALL
R-01	Reception	72	F1	W1
R-02	Office	72	F1	W1
R-03	Entrance hall	48	F1	W1
R-04	WC	48	F2	W2
R-05	Lockers	48	F2	W1
R-06	Electrical	120	F3	W3
R-07	Air plant	96	F3	W3
R-08	Maintenance	120	F3	W3
R-09	Tool crib	96	F3	W3
R-10	Returns	144	F3	W3

FLOOR FINISH INTENT

F1 Resilient staff finish.
F2 Wet-area or cleanable finish.
F3 Sealed service-grade concrete.

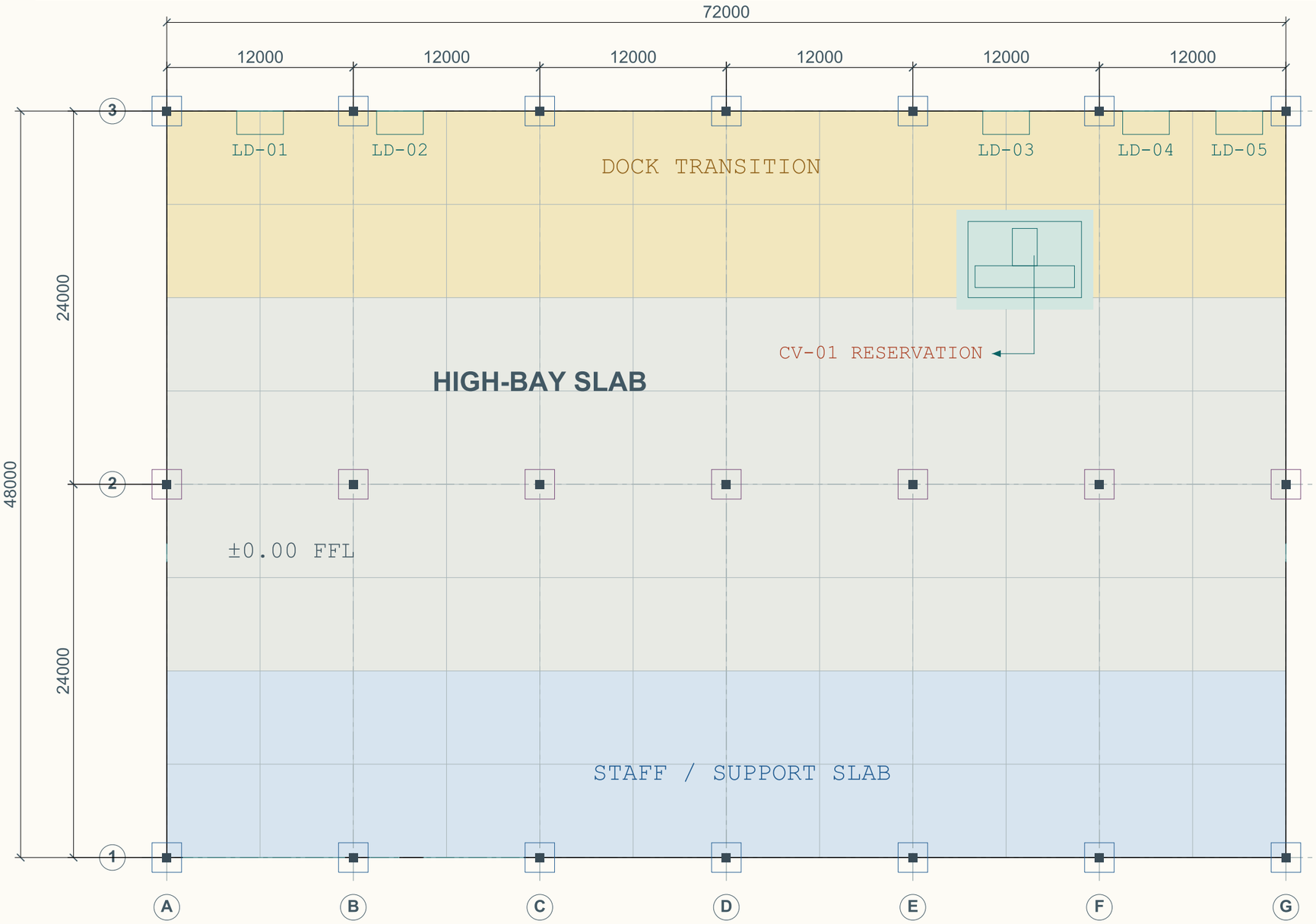
WALL FINISH INTENT

W1 Painted staff partition.
W2 Washable wet-zone lining.
W3 Sealed durable service wall.

*Gross plan allocations before wall deductions.

Foundations and slab coordination

COLUMN AND FOUNDATION CENTRES / SLAB ZONING / EQUIPMENT RESERVATIONS



01 FOUNDATION AND SLAB COORDINATION

SET-OUT REGISTER

LOCATION	CENTRES
Perimeter rows 1 + 3	14
Valley row 2	7
Total column points	21

FOUNDATION CONCEPT

Centres follow portal grid.
Pad and pedestal arrangement:
see base interface on S301.

SLAB / JOINT HOLD

Confirm wheel loads and racks.
Joint lines are coordination only.
Resolve anchor exclusion locally.

VECTOR RESERVATION

Origin X52.00 / Y36.65 m.
Guard: X51.55 to 58.85 m,
Y36.00 to 40.90 m.
Equipment interface: P102.

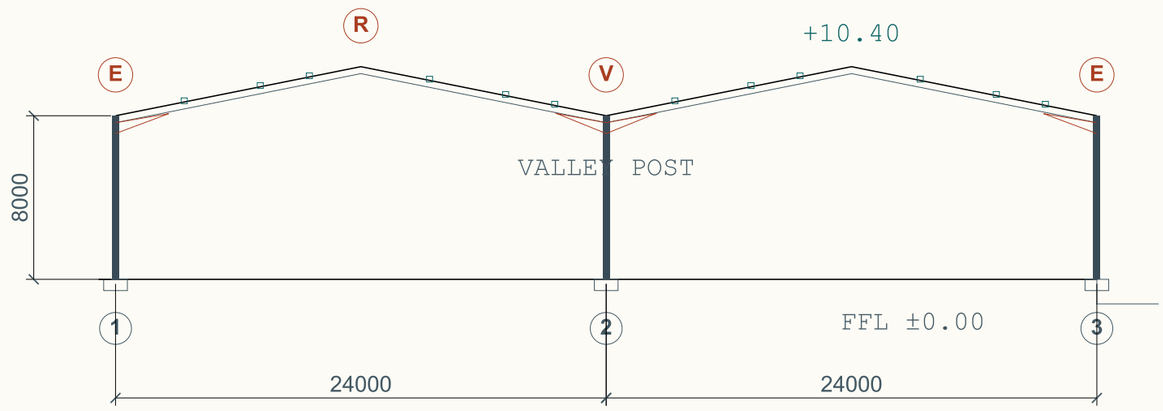
ENGINEERING INPUT

Geotechnical and reaction data.
Coordinate dock edge on A301.

Portal frames and stability

TWIN 24 M PORTALS / 12 M LONGITUDINAL BAYS / STABILITY LOAD PATH

01 TRANSVERSE PORTAL / GRID E



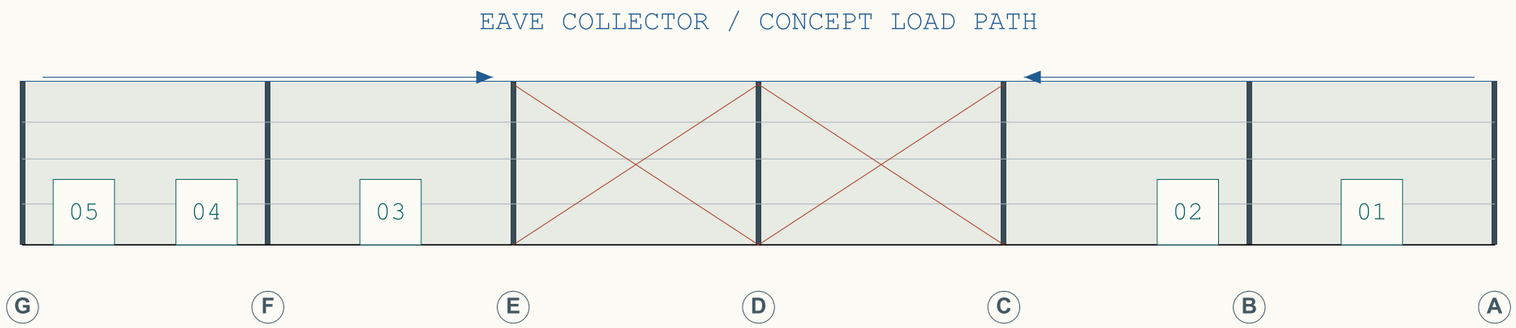
CONNECTION KEY

E Eave haunch / R Ridge splice
V Valley post node
Connection and base sketches on S301.

SECONDARY SUPPORT

Teal marks indicate purlin positions only.
Wall rails carry cladding to the columns.
12 m bay spacing requires span verification.

02 NORTH LONGITUDINAL STABILITY ELEVATION



STABILITY INTENT

Roof X-bracing: A–B / F–G.
Wall X-bracing: C–D / D–E.
Repeat on south long wall.
Verify collectors and forces.

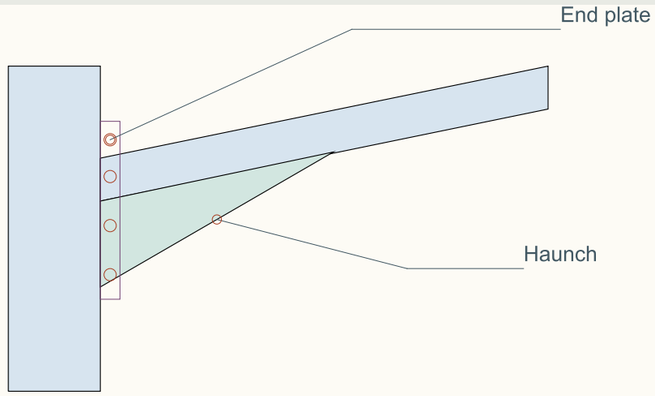
DOCK CHECK

LD-01 to LD-05 clear of proposed wall diagonals.

Structural interfaces

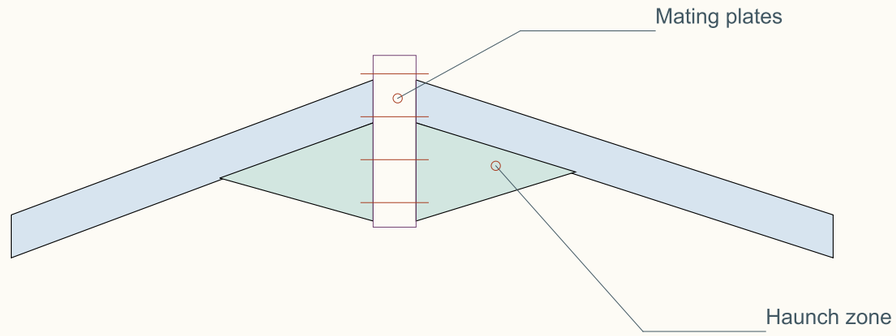
EAVE, RIDGE, VALLEY AND BASE / CONNECTION FAMILIES FOR DESIGN DEVELOPMENT

E EAVE HAUNCH



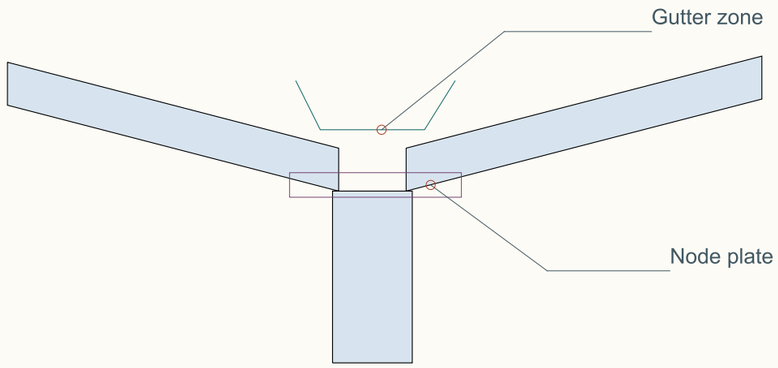
Moment connection geometry is indicative.

R RIDGE SPLICE



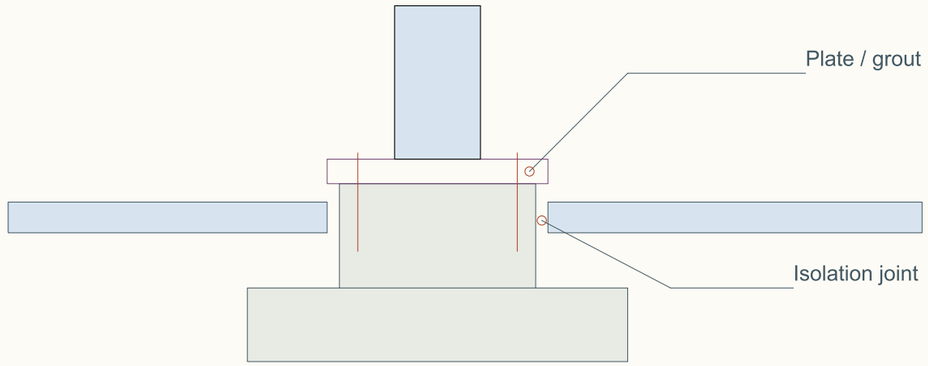
Splice and transport lengths to be coordinated.

V VALLEY POST NODE



Coordinate gutter support and service clearance on A501.

B PEDESTAL / SLAB INTERFACE

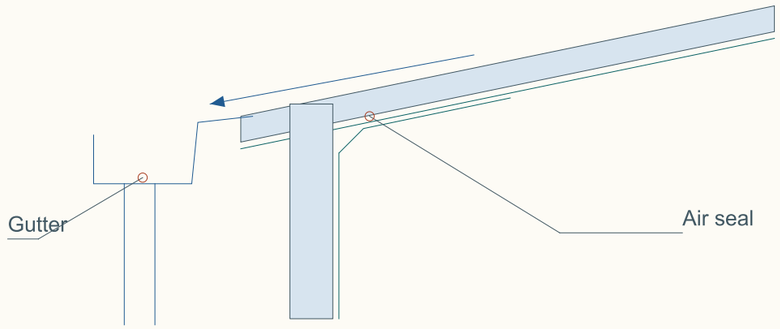


Footing and anchors require ground and reaction data.

Building envelope details

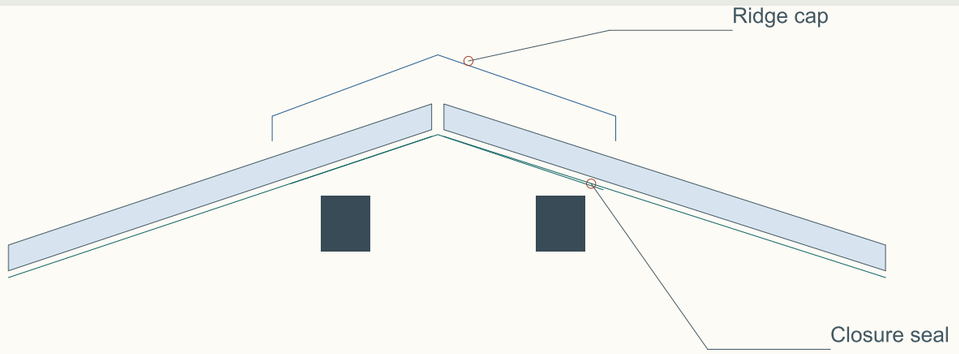
INSULATED ENVELOPE / RAINWATER AND AIR-SEAL CONTINUITY / TYPICAL INTERFACES

01 EAVE / OUTBOARD GUTTER



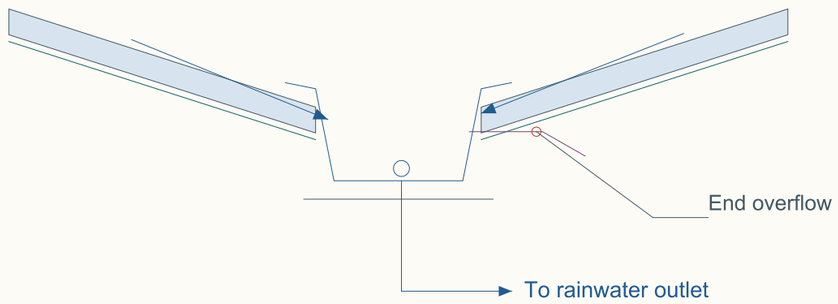
Downpipes at end-wall locations shown on S101 and C102.

02 RIDGE / CLOSURE



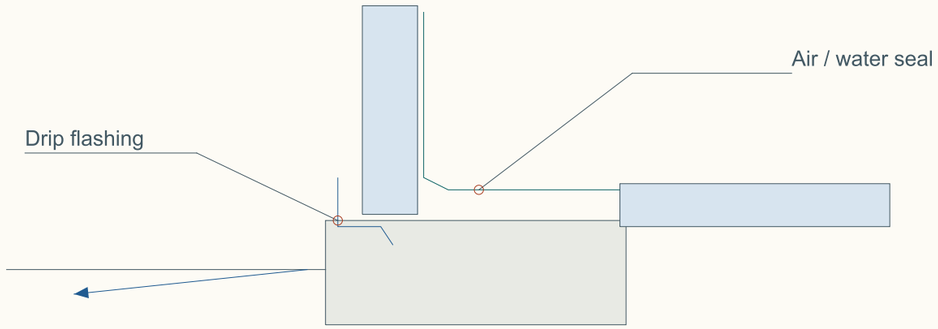
Carry the liner and air seal through the ridge transition.

03 VALLEY / END OVERFLOW



Verify hydraulic sizing, freeboard and visible overflow discharge.

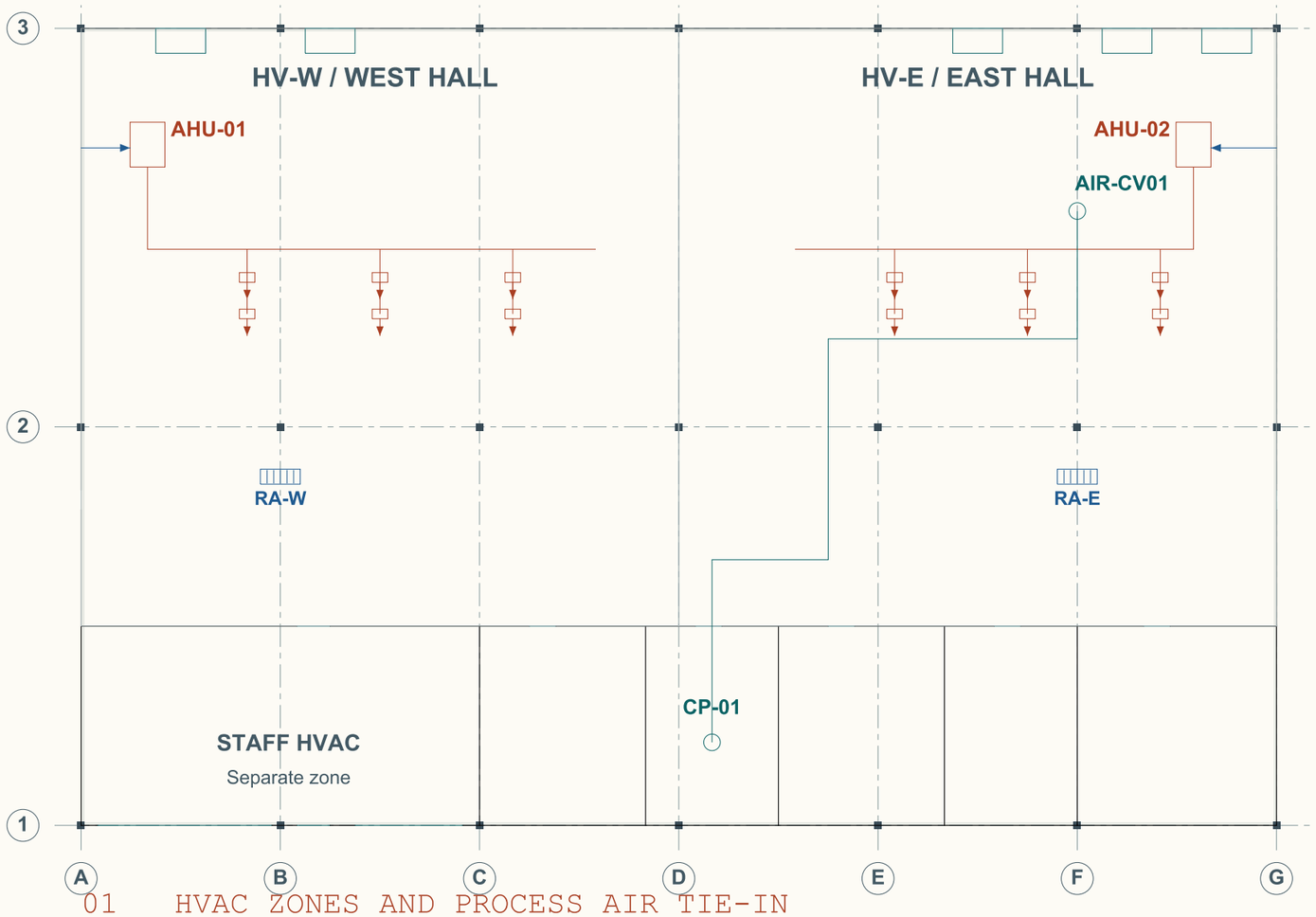
04 WALL / PLINTH TRANSITION



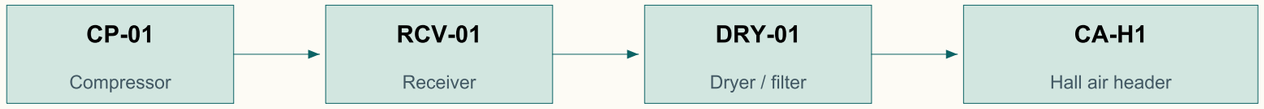
Verify grade, damp-proofing and cladding clearance to splash.

Air systems and elevations

VENTILATION INTENT / AIR-PLANT SCHEMATIC / SERVICE LEVELS



01 HVAC ZONES AND PROCESS AIR TIE-IN

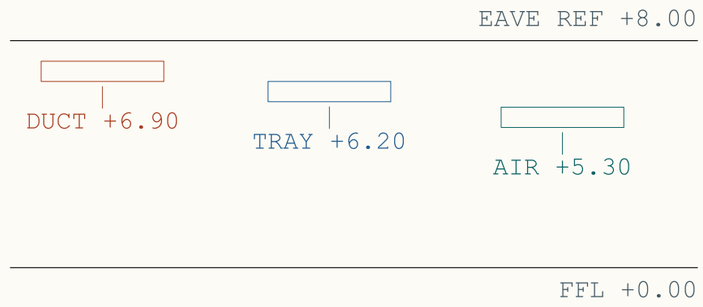


02 AIR PLANT FUNCTIONAL SEQUENCE / TREAT CONDENSATE SEPARATELY

VENTILATION ALLOCATION

TAG	SERVICE AREA	INTERFACE
AHU-01	West hall	West gable intake
AHU-02	East hall	East gable intake
RA-W / E	Hall return zones	Return / exhaust TBD
STAFF	Office + amenities	Dedicated HVAC
AIR-CV01	Vector transfer cell	Local isolation point

03 NOMINAL SERVICE SECTION



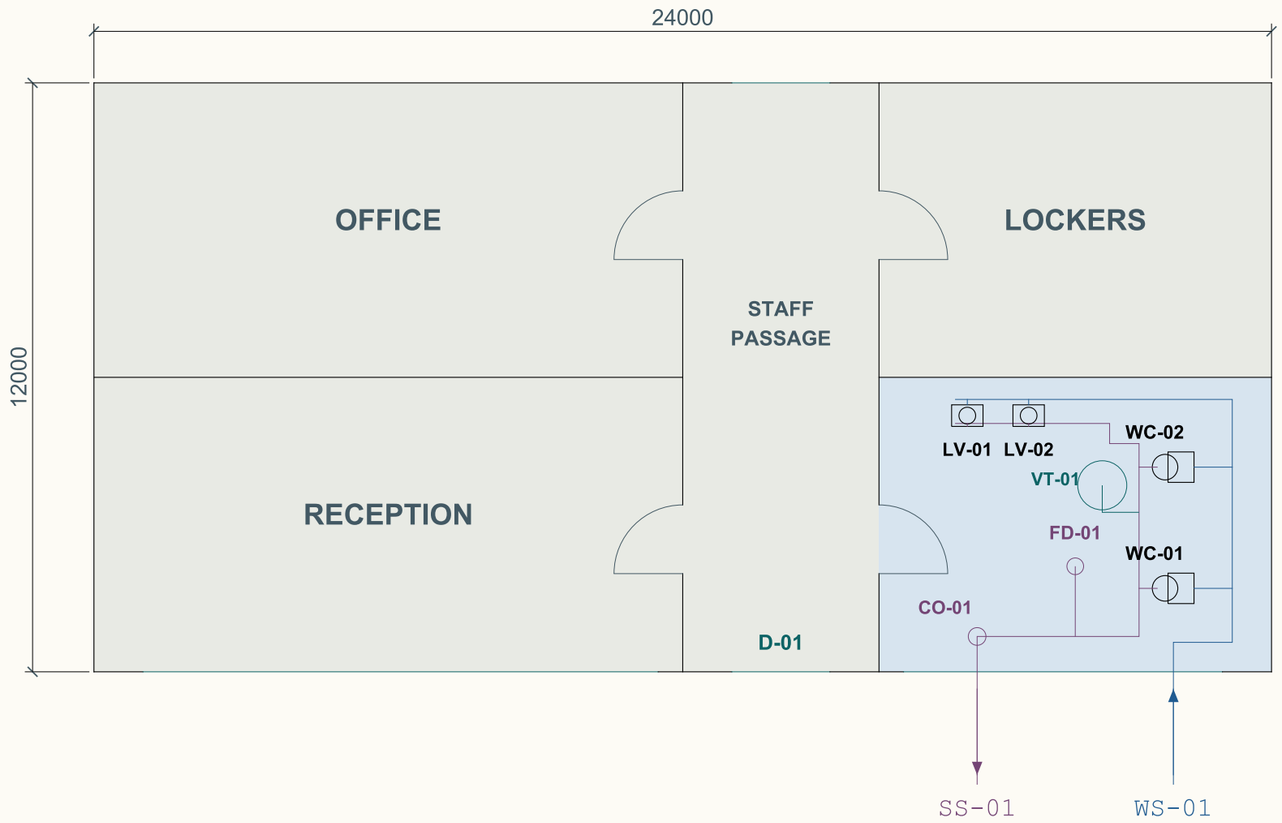
ENGINEERING INPUTS

Confirm heat gains, fresh air and extraction duty.
Select equipment and size ducts before resolving structure, sprinkler and access clearances.
Air capacity and pressure at CV-01 remain open.

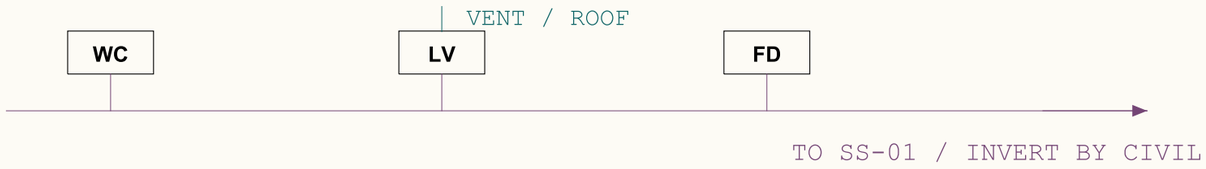
Water and sanitary drainage

STAFF CORE ENLARGEMENT / SEPARATE DOMESTIC AND SANITARY SITE INTERFACES

01 STAFF CORE / FIXTURE ROUTING



02 TYPICAL VENTED BRANCH / NTS



FIXTURE / ACCESS REGISTER

MARK	QTY	DESCRIPTION
WC-01 / 02	2	WC connection / cistern supply
LV-01 / 02	2	Wash basin / trapped waste
FD-01	1	Trapped floor drain / primer TBD
CO-01	1	Branch cleanout / maintenance access
VT-01	1	Vent continuation to roof

SITE CONNECTION ALLOWANCES

TAG	HALL / SITE X	INTENT
WS-01	22 m / 52 m	Domestic service
SS-01	18 m / 48 m	Sanitary discharge

ROUTE KEY

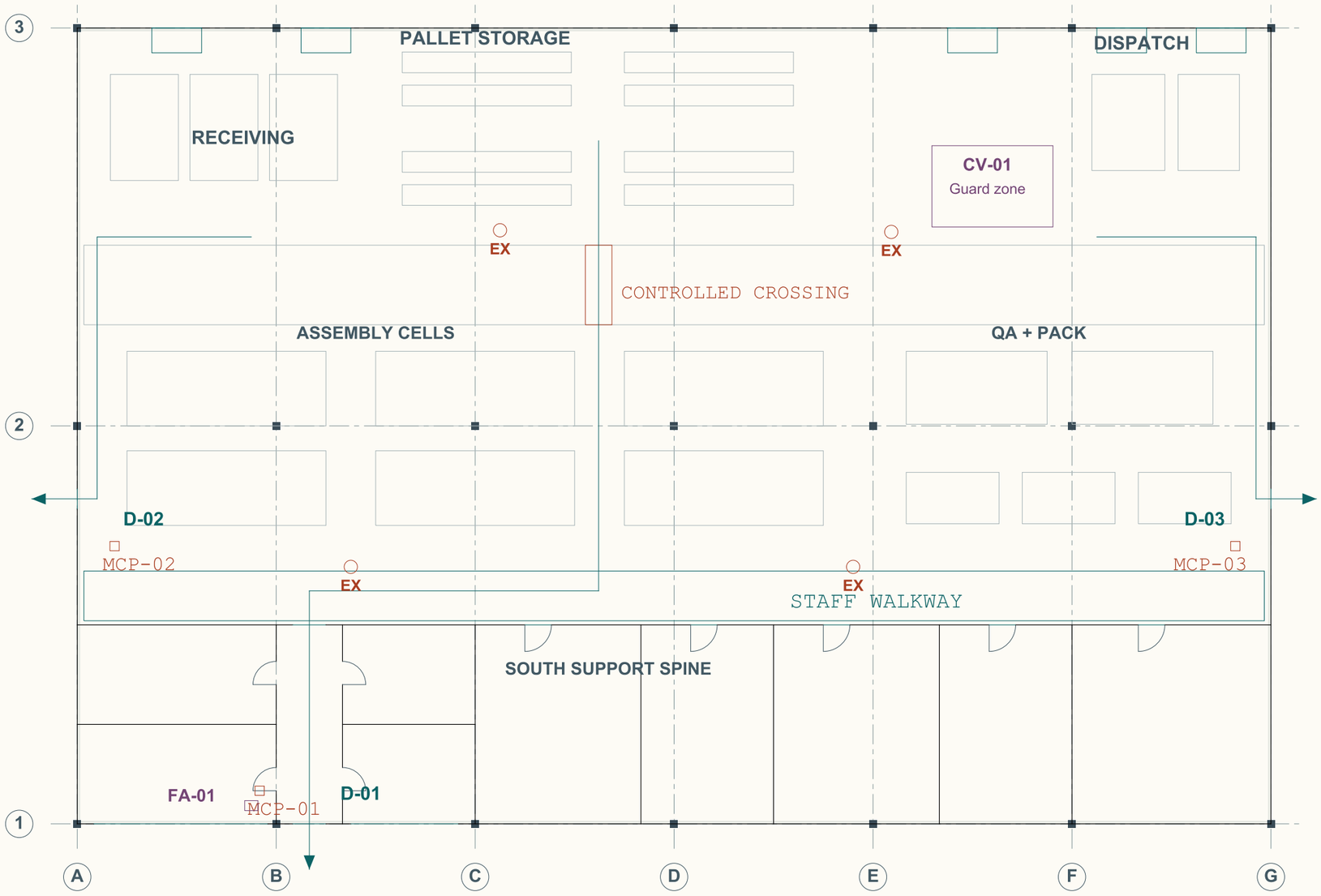
Blue / domestic water
Plum / sanitary drainage
Teal / vent continuation
Roof and yard water remain separate on C102.

RELEASE INPUTS

Confirm fixture counts, accessibility, supply pressure, vent terminal and sewer invert.

Egress and fire access

INDICATIVE EXIT ACCESS / EXIT DISCHARGE / FIRE ALARM AND EQUIPMENT LOCATIONS



01 EXIT ACCESS AND DISCHARGE INTERFACES

EXIT INTERFACE REGISTER

MARK	FACE	OUTSIDE
D-01	South	Staff approach
D-02	West	West bypass
D-03	East	East bypass

PLAN KEY

Teal arrow / indicative exit route
MCP / manual alarm point
EX / extinguisher allowance
FA-01 / fire alarm panel

EXIT DISCHARGE

Follow civil pedestrian routing to AP-01, west of the entry.
Site coordinate: (24, 13) m.
Keep truck apron off exit routes.

RELEASE INPUTS

Confirm occupancy and local rules.
Verify exit count, clear widths, travel, doors and lighting.

RESPONSE ACCESS

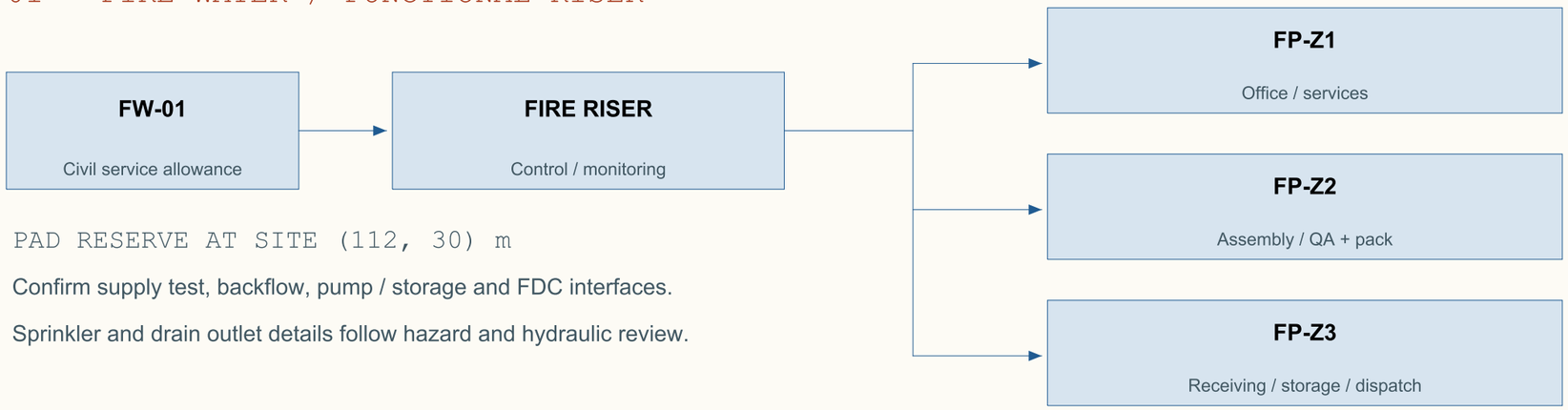
East-side fire-water reserve FW-01 on C102 / F201.
Confirm appliance access.

N

Fire system interfaces

SUPPRESSION ZONES / WATER ENTRY / ALARM AND EQUIPMENT INTERFACE MATRIX

01 FIRE WATER / FUNCTIONAL RISER

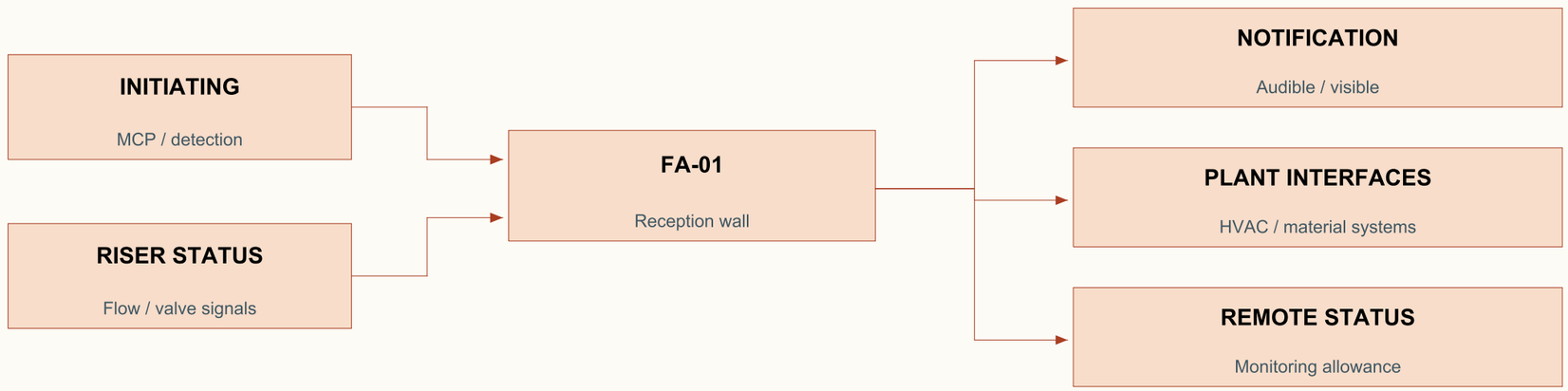


PAD RESERVE AT SITE (112, 30) m

Confirm supply test, backflow, pump / storage and FDC interfaces.

Sprinkler and drain outlet details follow hazard and hydraulic review.

02 FIRE ALARM / SIGNAL INTERFACES



ALARM ACTIONS REQUIRE AN APPROVED CAUSE-AND-EFFECT MATRIX.

ZONING BASIS

GROUP	EXTENT
FP-Z1	South staff / services
FP-Z2	Assembly and QA + pack
FP-Z3	North logistics + storage

SIGNAL HANDOFF / INTENT

INPUT	REVIEW POINT
Manual / detect	Notify / plant interface
Waterflow	Alarm review
Valve status	Supervisory review

STORAGE INPUTS

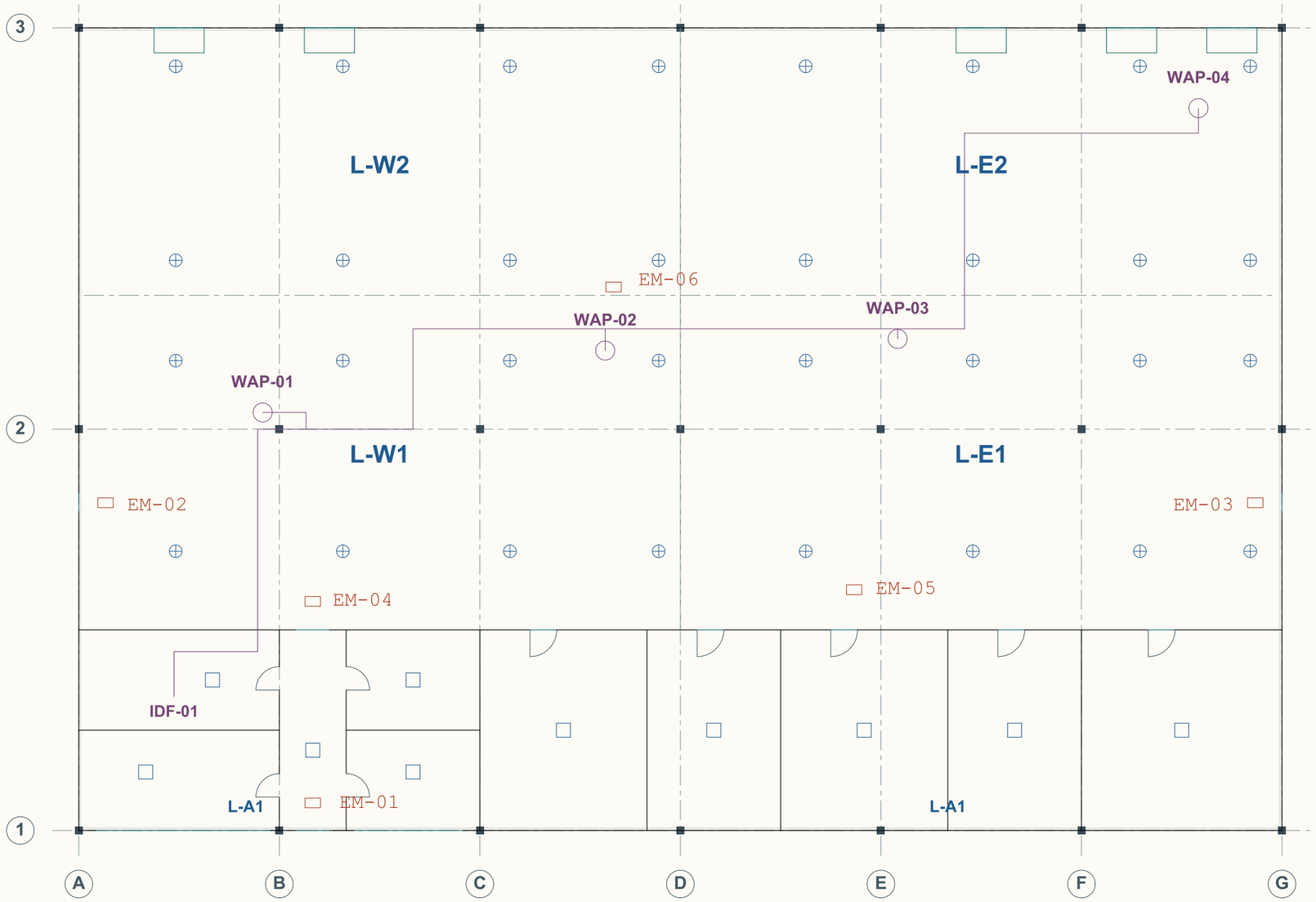
Confirm commodity and packaging, racks, heights and obstructions. Determine ceiling / in-rack basis.

SYSTEM RELEASE

Confirm fire-water flow / pressure, hydraulics, alarm zoning and cause-and-effect with the AHJ. Exit and access locations on F101.

Lighting and low voltage

LIGHTING CONTROL GROUPS / EMERGENCY LOCATIONS / DATA AND WI-FI RESERVES



01 LIGHTING, EMERGENCY AND ELV LAYOUT

LIGHTING GROUP REGISTER

GROUP	CONTROL AREA
L-W1 / E1	West / east workcells
L-W2 / E2	West / east logistics
L-A1	Staff / support rooms
EM-01–06	Egress allowances

SYMBOL KEY

Blue circle / high-bay light
Blue square / staff light
Rust rectangle / emergency
Plum circle / wireless AP

CONTROL INTENT

Local occupancy / zone switching.
Confirm task and emergency lux.
Select mounting heights, circuits and emergency power strategy.

ELV INTERFACE

IDF-01 to WAP-01 through WAP-04.
Survey coverage and cable route.
Panel allocations on E602.

N

Distribution and bonding

BOARD SCHEDULE / CONTROL HANDOFF / PROTECTIVE BONDING INTERFACES

01 DISTRIBUTION AND LOAD GROUPS

BOARD	FEED	LOAD / FUNCTION	REFERENCE
MSB-01	Q0	Utility service; electrical room	E601 / M101
DB-PROD	Q1	AC-01–06, QA-01–02, PK-01–03	P101 / M101
DB-LOG	Q2	LD-01–05, rack / dispatch, CV-01	M101 / (65, 36.9) m
DB-ADM	Q3	Staff loads, L-A1, IDF-01, FA-01 allowance	E201 / F201
MCC-01	Q4	CP-01, AHU-01, AHU-02	M201 / E601
L-W1 / E1	TBD	High-bay workcell zones	E201
L-W2 / E2	TBD	High-bay logistics zones	E201
EM-01–06	TBD	Emergency lighting supply / backup	E201 / F101

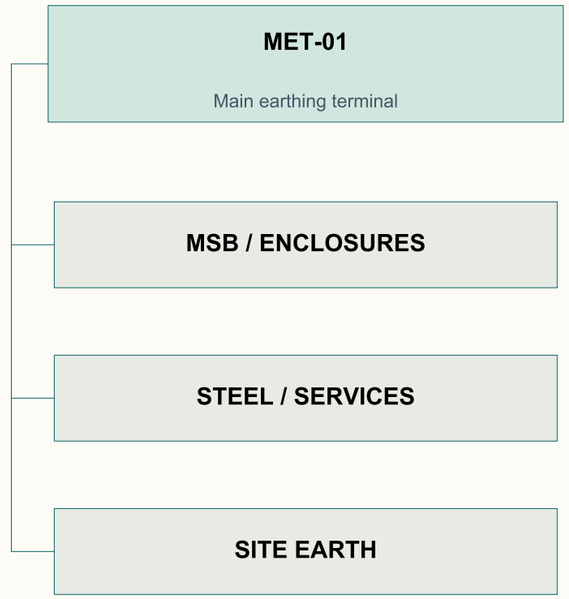
02 VECTOR CV-01 / BUILDING INTERFACES

INTERFACE	BUILDING SIDE	EQUIPMENT SIDE / ACTION
PWR-CV01	DB-LOG feeder allowance	Vendor disconnect / verify service, load and protection
NET-CV01	ELV route reservation	Controls network and field I/O by equipment package
AIR-CV01	Local stub at (60, 37) m	Local isolation; confirm air quality, pressure and flow
ALM-CV01	FA-01 interface allowance	Approve cause-and-effect and safety boundaries

LOAD, FEEDER, FAULT AND PROTECTION STUDIES FOLLOW VERIFIED SERVICE AND EQUIPMENT DATA.

CV-01 field wiring belongs to the equipment drawing package.

03 PROTECTIVE BONDING



COORDINATION HOLD

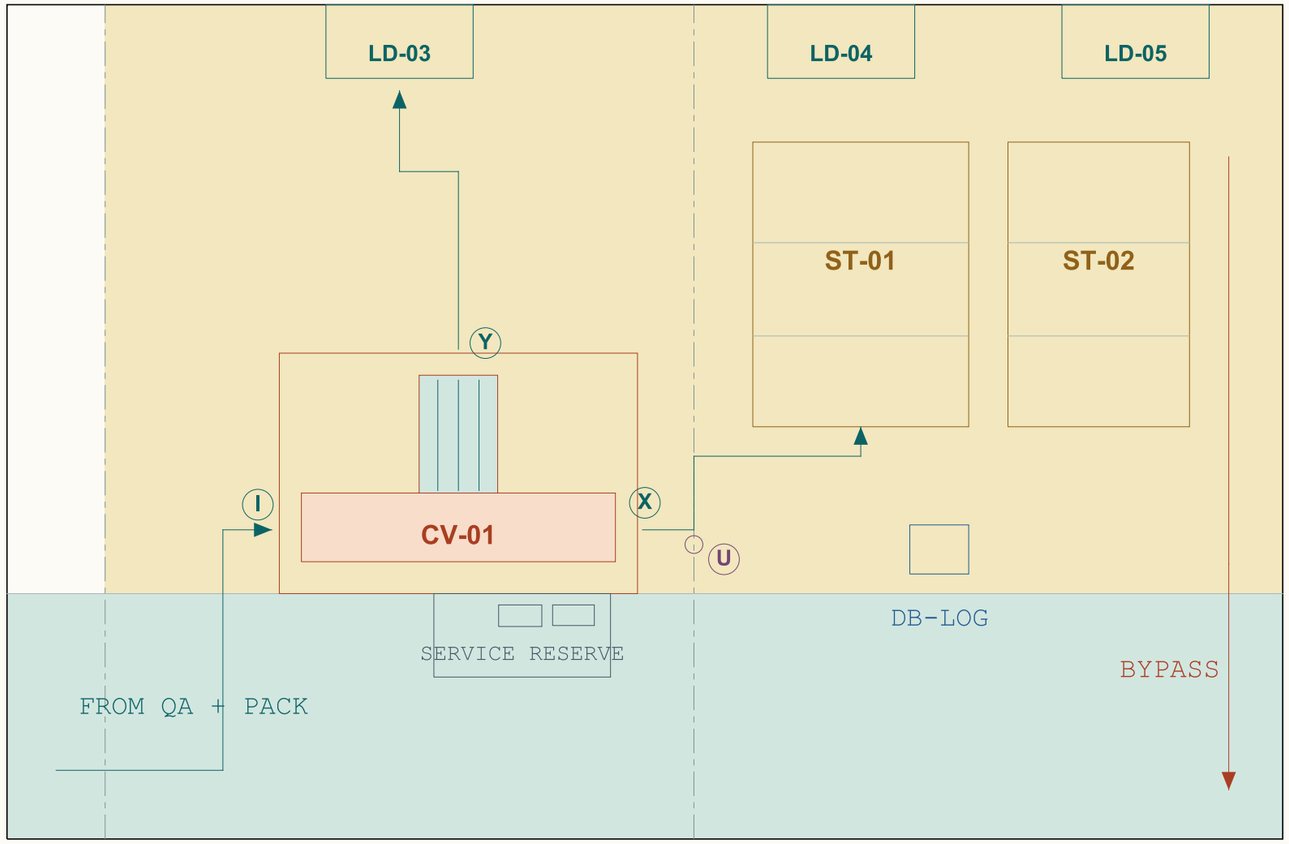
Bonding relationships only. Confirm service earthing scheme, electrode and conductor sizing, earth continuity and test points.

DESIGN BASIS

Service characteristics and equipment ratings remain open.

Vector equipment integration

DISPATCH ZONE ENLARGEMENT / CV-01 FROM VECTOR EQUIPMENT SET REV 01



01 ENLARGED DISPATCH INTERFACE

SET-OUT BASIS

Hall-local equipment origin X=52.00 / Y=36.65 m.
Equipment origin in site coordinates: X=82.00 / Y=58.65 m.

Guard envelope X 51.55–58.85 / Y 36.00–40.90 m.

HANDOFF POINTS

ID	INTERFACE	COORDINATION
I	Infeed from QA + pack	West material interface
X	Straight discharge	East staging / LD-04–05
Y	Side discharge	North to LD-03
U	Utilities drop / (60,37)	DB-LOG / air header

EQUIPMENT REGISTER

CV-01 assembly envelope 6.4 × 3.8 m.
Roller top +0.90 m above hall floor.
Side deck +0.93 m above hall floor.

RELATED DRAWINGS

M101 service route / E602 board allocation.
F101 egress map / S102 anchorage interface.
Vector Rev 01 G000, M120 and M400.

RELEASE INPUTS

Validate handling swept paths and guarding.
Confirm anchors, headroom and supply ratings.
Reserve the marked bypass during installation.

Coordination and acceptance

CROSS-DRAWING COORDINATION / OPEN ENGINEERING INPUTS / ACCEPTANCE RECORDS

01 DRAWING COORDINATION REGISTER

ID	INTERFACE	COMMON BASIS	CHECK DRAWINGS
01	Site / hall datums	Hall local origin maps to site (30,22) m.	G000 / C101 / C102 / A101
02	North loading front	Five docks, 3.0 m clear width, apron −1.20 m.	A101 / A201 / A301 / A601
03	Portal / roof	2 × 24 m, 6 × 12 m, +8.00 / +10.40 m.	S101 / S201 / S301 / A501
04	Drainage separation	Roof to west rain garden. Yard treatment and outlet to confirm.	C102 / S101 / A501
05	Equipment integration	CV-01 X52.00 / Y36.65, feeds via DB-LOG and air.	P101 / P102 / S102 / E602
06	Life safety interfaces	Three hall personnel exits, keep site discharge clear.	A101 / A202 / F101 / F201
07	Utilities / structure	Coordinate route depth, offsets and maintenance clearance.	M101 / M201 / S201

02 ENGINEERING INPUTS AND ACCEPTANCE

SITE + CIVIL
Survey, invert/outfall and storm criteria.
Vehicle swept path and apron grading.
Record site datum before set-out.

STRUCTURE + EQUIPMENT
Geotechnical and load data.
Check member, slab and anchor capacity.
Verify delivery route and machinery trials.

FIRE + MEP
Local requirements, occupancy and storage.
Confirm utility capacity and calculations.
Approve alarm, lighting and commissioning tests.

DOCUMENT CHECK
Compare sheet index and issue revision.
Verify model units, layouts and dimension objects.

INTERFACE CHECK
Trace each shared tag to its source sheet.
Compare roof, dock, exit and equipment coordinates.

HOLD POINT
Concept coordination issue.
Engineering inputs support the next issue.

Development issue register

Q001 FOLLOW-UP / 11-SHEET SUPPLEMENT / USE WITH REV 01 BASELINE SET

01 DEVELOPMENT DRAWING REGISTER

ID	DRAWING	REVIEW BASIS	RELEASE DEPENDENCY
Q002	Development issue register	REV 01 Q001	Maintain agreed input status and issue history.
C201	Civil setting-out development	C101 / C102	IN-02 survey and CAL-01 grading, vehicle paths and pavement.
C501	Stormwater installation interfaces	C102 / A301 / A501	IN-02/05 outfall, invert and storm criteria. CAL-01 hydraulics.
A302	Dock installation interfaces	A301 / A601	IN-04 approved dock vendor pit and interface reactions.
S103	Foundation grid set-out	S101 / S102	IN-02/03 survey, geotechnical data and reactions. CAL-03.
S202	Steel frame coordination	S201 / S301	IN-03 loads and CAL-03 member / connection checks.
S302	Slab equipment and dock interfaces	P102 / S102 / A301	IN-03/04 slab loads, equipment anchors and pit design.
M301	Overhead service installation	M101 / M201	IN-04/05 selected sizes. CAL-04/05 supports and clearances.
F301	Fire protection installation interfaces	F101 / F201	IN-01/02/05 storage, authority criteria and fire-water data.
E603	Electrical room and field interfaces	E601 / E602	IN-04/05 loads, supply and fault data. CAL-06 coordination.
Q003	Geometry and runoff study checks	G000 / C102 / S101	Trial rainfall values for study. Design storm remains open.

02 INTERPRETATION OF THIS ISSUE

REV 01 BASIS

Modelled geometry and tags.
Survey and authority checks remain open.

PORTFOLIO ASSUMPTION

Trial inputs identified by name.
Recalculate when actual data becomes available.

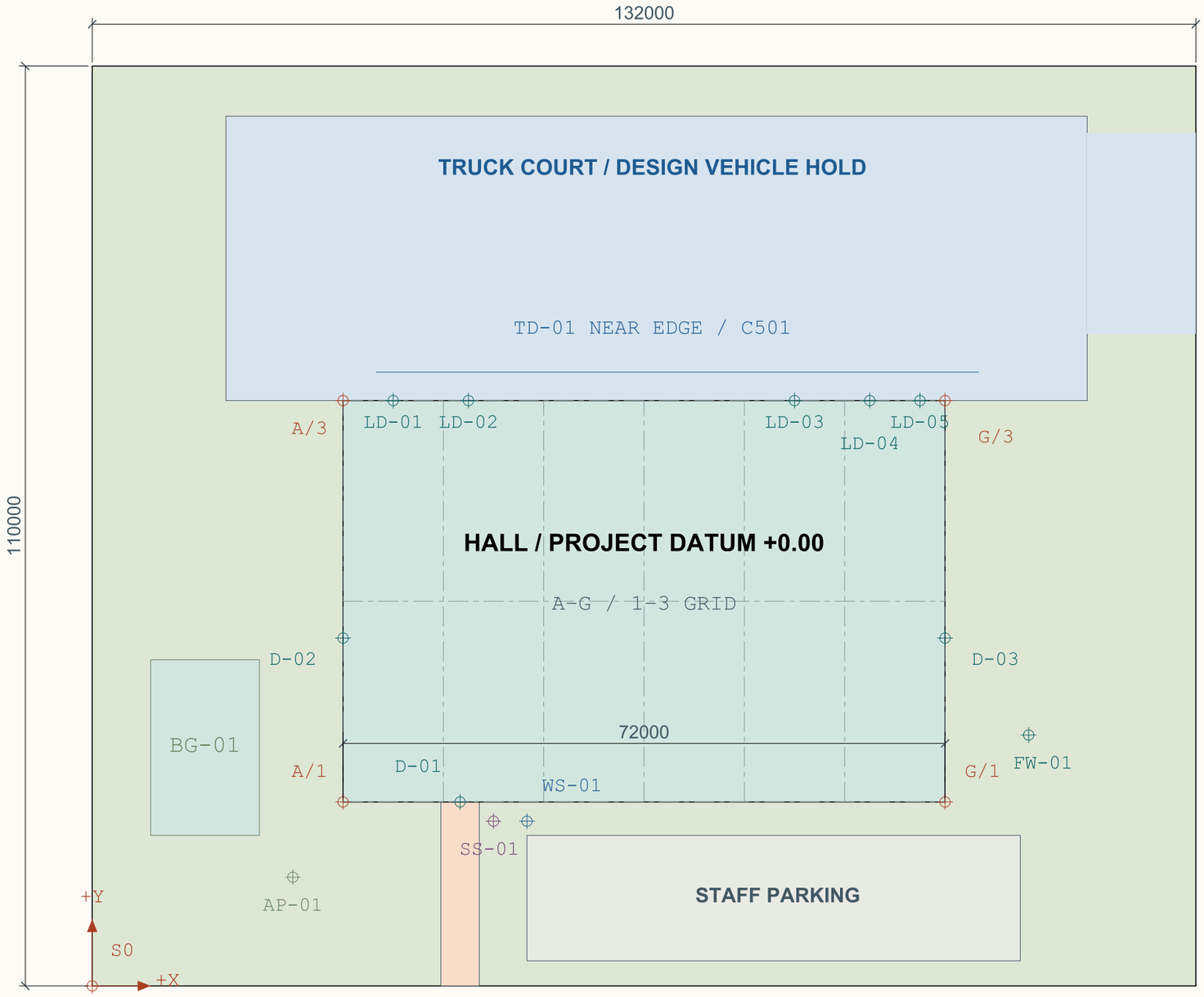
HOLD

Sizing, drilling and fabrication depend on the released design and vendor data.

Input and calculation IDs refer to the pre-drafting Q001 development register.

Civil setting-out development

COORDINATE SCHEDULE / GRADING CONTROL POINTS / BENCHMARK AND VEHICLE HOLD



SITE SET-OUT / METRES

TAG	X	Y	REF
A/1	30.00	22.00	Hall origin
G/3	102.00	70.00	Opposite corner
LD-01	36.00	70.00	Dock centre
LD-02	45.00	70.00	Dock centre
LD-03	84.00	70.00	Dock centre
LD-04	93.00	70.00	Dock centre
LD-05	99.00	70.00	Dock centre
D-01	44.00	22.00	Entry centre
D-02	30.00	41.60	Entry centre
D-03	102.00	41.60	Entry centre
SS-01	48.00	19.70	Sanitary stub
WS-01	52.00	19.70	Water stub
AP-01	24.00	13.00	Assembly point
FW-01	112.00	30.00	Fire water reserve

GRADING COORDINATION

C102 relative levels retained. No surveyed RL.
North dock −1.20 / South approach −0.25.
East gate −0.70 / Yard capture −1.25.

RELEASE HOLD / SURVEY

Tie S0 and FFL to approved survey benchmark.
Confirm boundaries, swept paths and utilities.

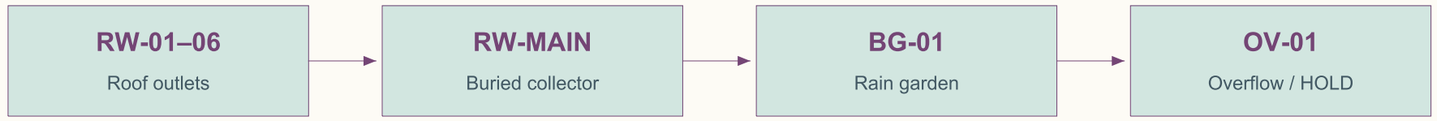
0 10 20 30 m

N

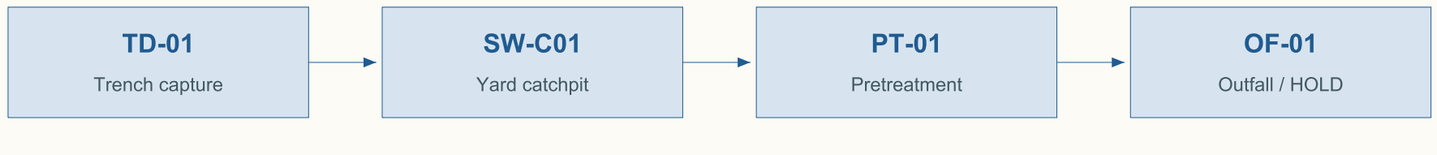
Stormwater installation interfaces

ROOF WATER AND TRUCK-YARD SYSTEMS / SEPARATE FLOW PATHS / INVERT RELEASE HOLD

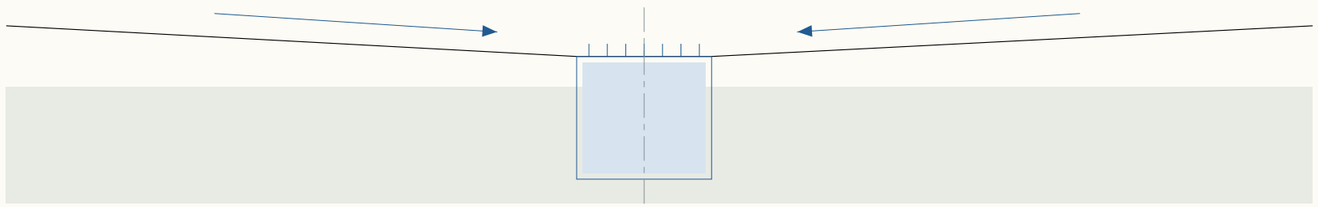
01 CLEAN ROOF-WATER SYSTEM



02 TRUCK-YARD RUNOFF SYSTEM



03 TD-01 / TYPICAL CROSS-SECTION NTS



OUTLET / INVERT HOLD

Grate class, bedding, edge restraint and backfill require vendor / civil review.

Drainage surface from C102 is an illustrative grading control, subject to survey.

TD-01 SITE OFFSET CHECK

REFERENCE	SITE Y (M)
North dock face	70.00
A301 trench near edge	73.400
A301 indicative centre	73.575
C102 route reference	73.400

RECONCILIATION / HOLD

C102 shows the route at the near edge.
A301 shows a 0.35 m indicative band.
Confirm vendor width and signed set-out.

HYDRAULIC RELEASE

Confirm storm criteria, invert and outfall.
Size capture, treatment and overflow.
Verify rain-garden soil and infiltration.

KEEP SYSTEMS SEPARATE

Yard water requires agreed treatment.
WS-01 and SS-01 stay separate.
Coordinate crossings with P201 / C201.

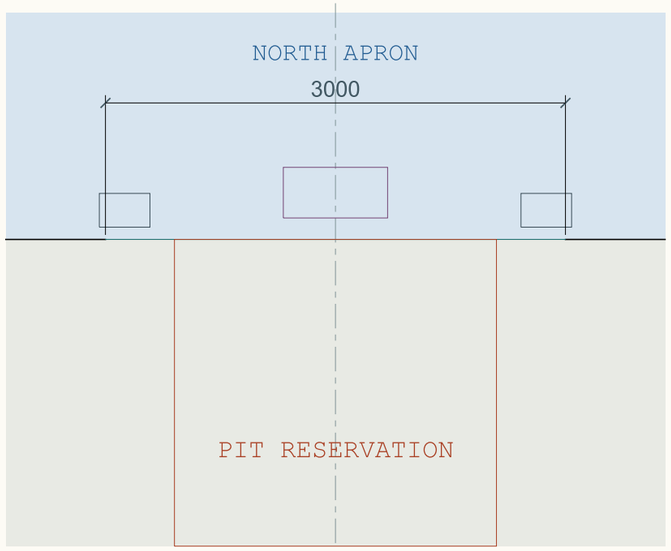
SECTION LINKS

A301 / A302 for dock and pit interfaces.

Dock installation interfaces

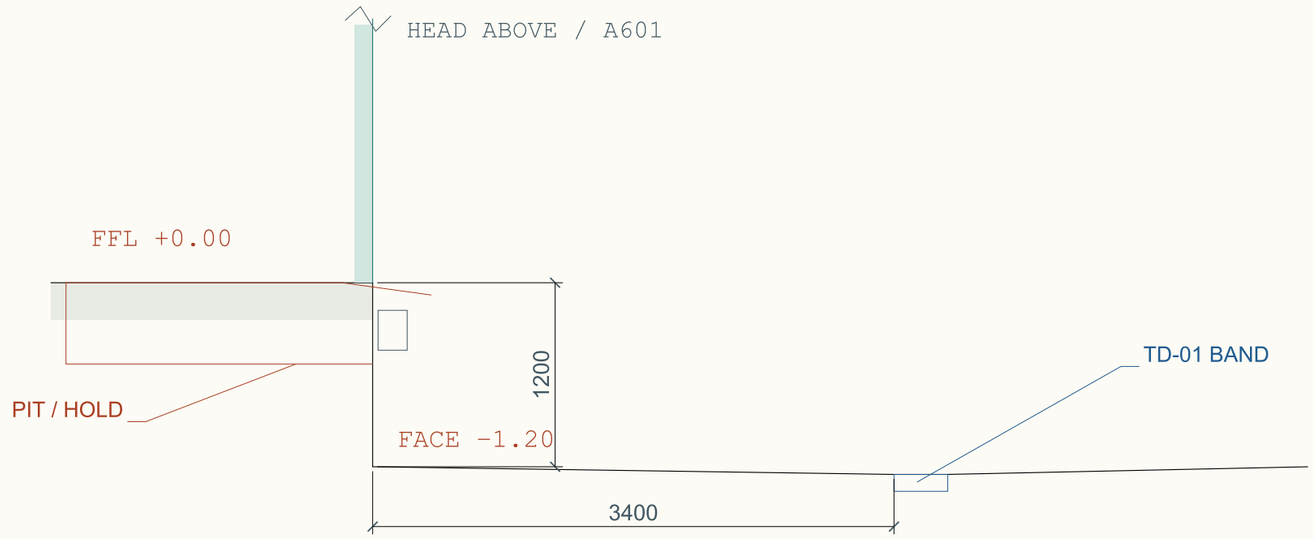
NORTH DOCK / DOOR AND LEVELLER GEOMETRY / VENDOR AND STRUCTURE RELEASE HOLD

01 TYPICAL DOCK / PLAN



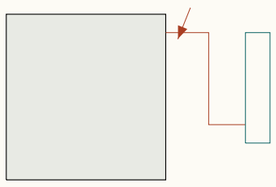
PIT LENGTH, WIDTH AND DEPTH / VENDOR HOLD

02 TYPICAL SECTION / DOOR CENTRE



3,400 TO TD-01 NEAR EDGE PER A301 / VERIFY VENDOR SET-OUT

03 DOOR JAMB / NTS



INTERFACE
Seal / flashing and track.
Jamb support / anchorage.
Confirm fixing with vendor.

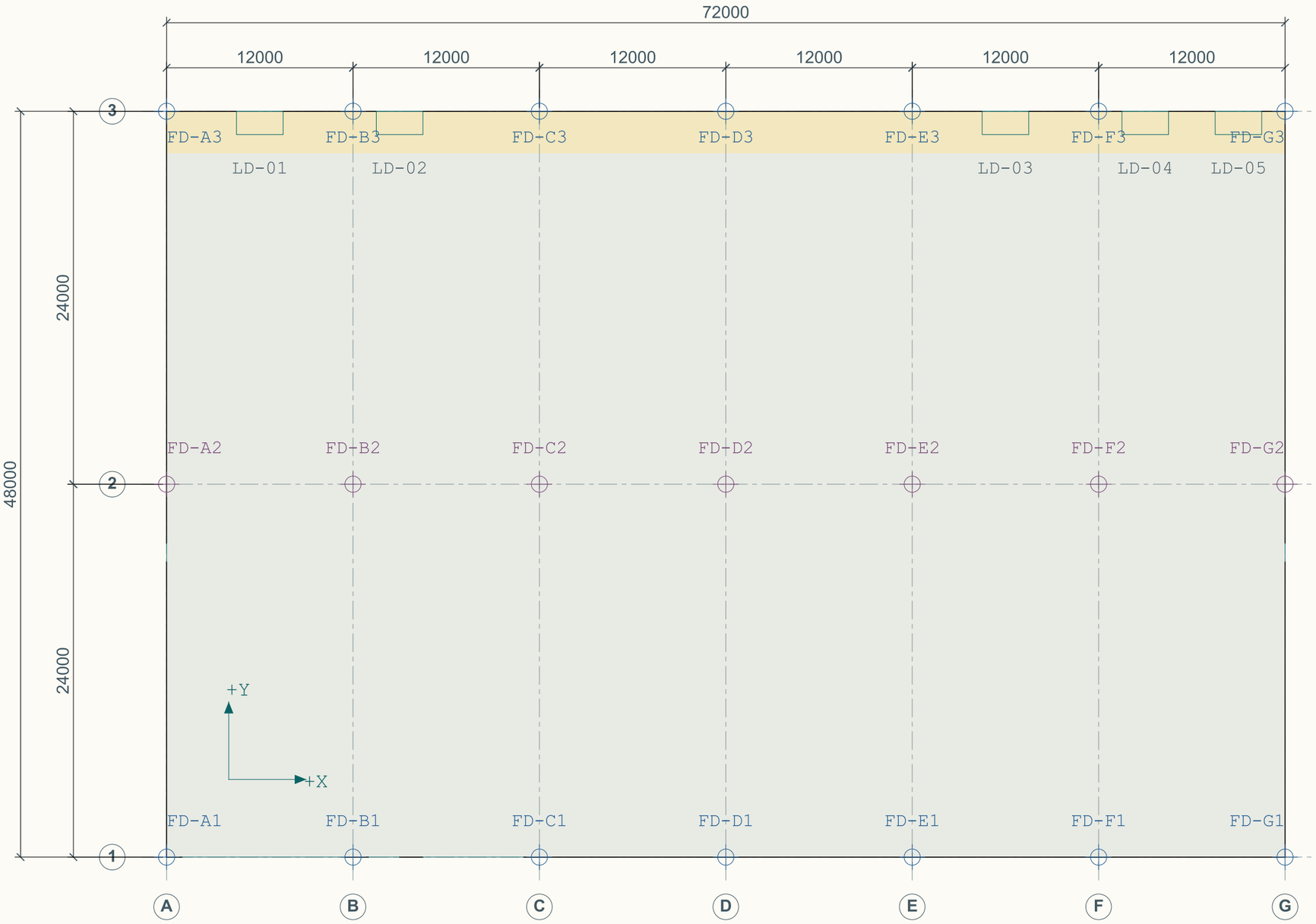
A501 / A601 / S301

04 DOCK SUBMITTAL / RELEASE REGISTER

ID	COORDINATION REQUIREMENT	RELEASE OWNER
H-01	Dock doors: 3,000 clear × 3,200 clear	Architect / door supplier
H-02	Leveller pit, embed plates and reactions	Dock vendor / structure
H-03	Restraint, bumpers and impact loads	Dock vendor / structure
H-04	Trench section, invert and grated load class	Civil / drain supplier
H-05	Interlock, controls and utility rough-ins	Electrical / dock vendor

Foundation grid set-out

FOUNDATION CENTRES / HALL-TO-SITE TRANSFORM / REACTION AND FOUNDATION RELEASE



01 FOUNDATION GRID / CENTRE MARKS

X GRID REGISTER / M

GRID	HALL X	SITE X
A	0.00	30.00
B	12.00	42.00
C	24.00	54.00
D	36.00	66.00
E	48.00	78.00
F	60.00	90.00
G	72.00	102.00

Y GRID REGISTER / M

GRID	HALL Y	SITE Y
1	0.00	22.00
2	24.00	46.00
3	48.00	70.00

SET-OUT BASIS

FD-B2 example: hall (12,24).
Site coordinate (42,46).
21 provisional column centres.

FOUNDATION RELEASE HOLD

Survey benchmark and site tie.
Geotechnical profile and water.
Engineer reactions and sizing.
No footing extent is indicated.

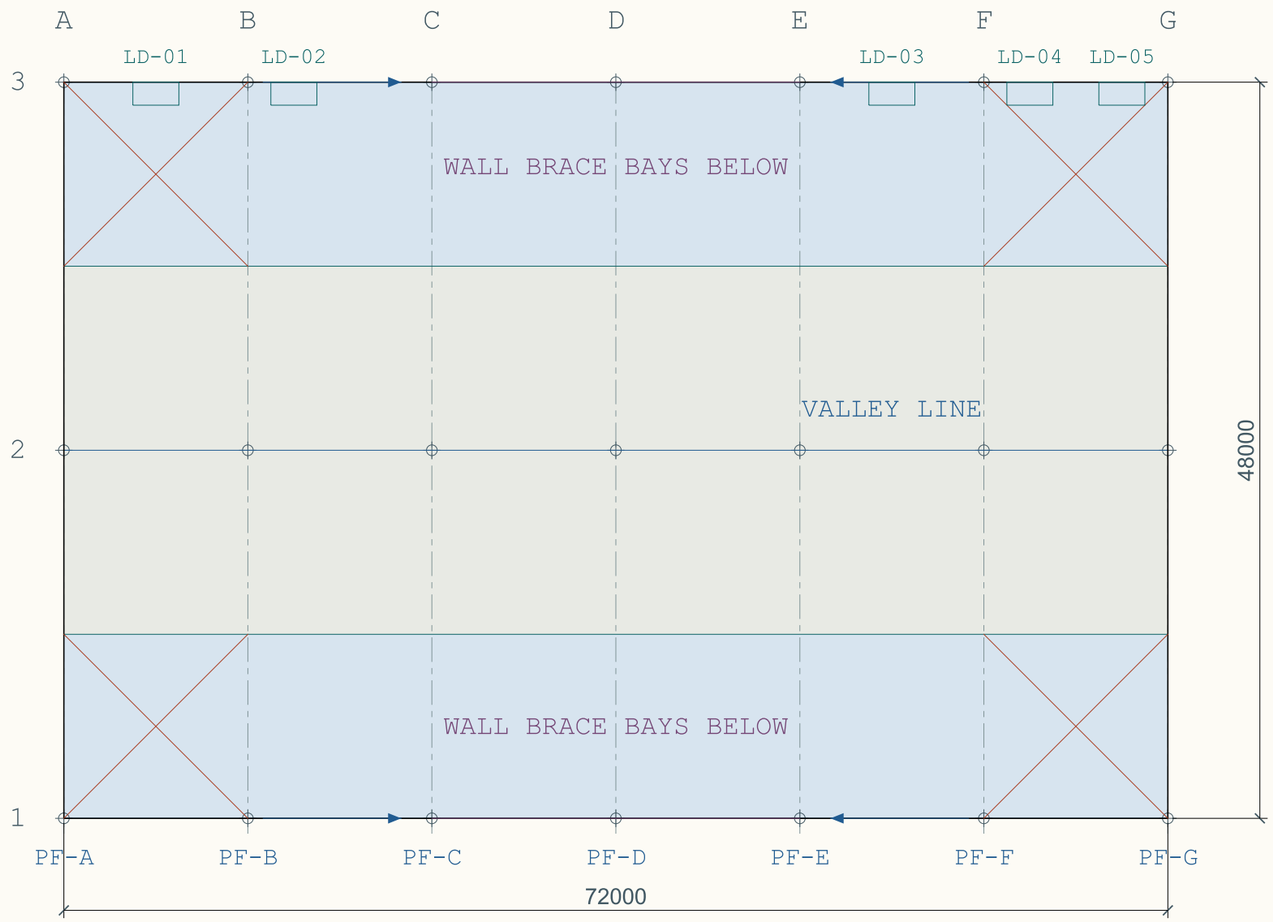
INTERFACES

C102 / S102 / S202 / A301

Steel frame coordination

PORTAL AND STABILITY SET-OUT / CONNECTION FAMILY REGISTER / MEMBER RELEASE

01 FRAME LINES AND STABILITY ZONES



FRAME ID CONTRACT

PF-A to PF-G locate transverse portals.
Use FD-grid references for reactions.
IDs identify location, not member equivalence.

STABILITY INTERFACES

Roof X braces: end bays A-B and F-G.
Wall X brace zones: C-D and D-E, rows 1 and 3.
Eave collectors link the two systems.
S201 north view reads G to A.

CHECKS BEFORE RELEASE

Load combinations and second-order stability.
Restraints and 12 m secondary spans.
Movement, deflection, connections and uplift.
Rack, PV, MEP and suspended loads.

NO MEMBER SECTIONS ASSIGNED

Engineer member, joint and base schedules.
Coordinate fire, corrosion and erection.

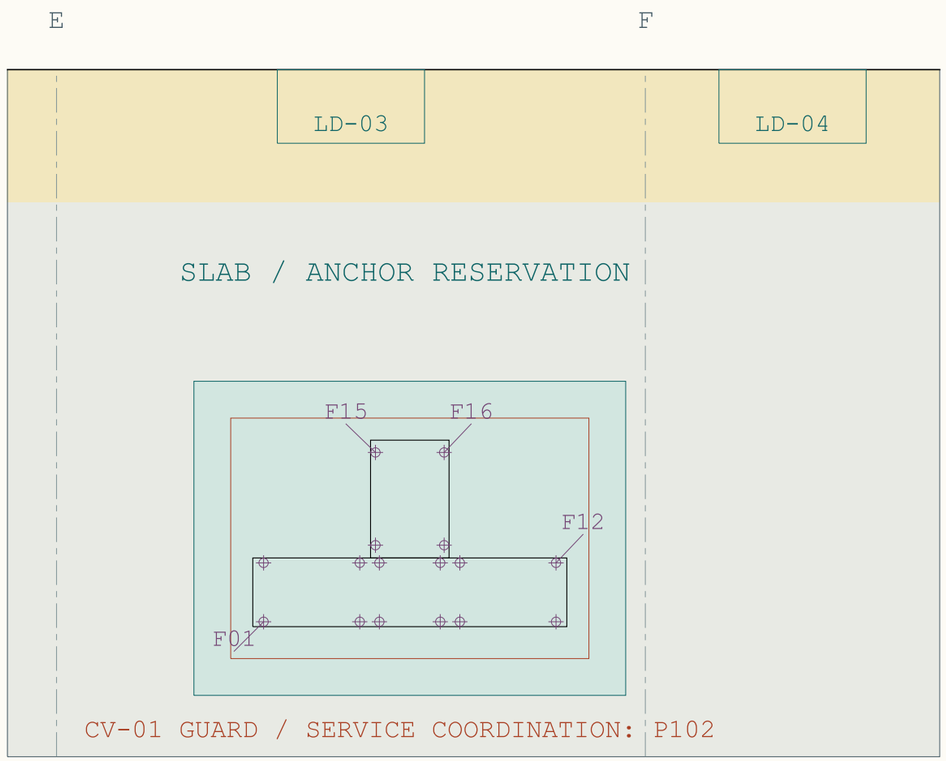
02 CONNECTION FAMILY / RELEASE REGISTER

FAMILY	LOCATION / SOURCE	RELEASE INPUT
E / R / V	Eave, ridge and valley / S301	Frame actions and joint stiffness basis before bolts, welds and plates.
BR / CL	Bracing and collectors / S201	Force path, attachments, restraint and access clearances.
FD	Column bases / S103	N, V, M and uplift by combination; verify anchor and concrete resistance.

Slab equipment and dock interfaces

DISPATCH SLAB / CV-01 CANDIDATE FOOT CENTRES / NORTH DOCK EDGE

01 DISPATCH INTERFACE / CANDIDATE FOOT CENTRES



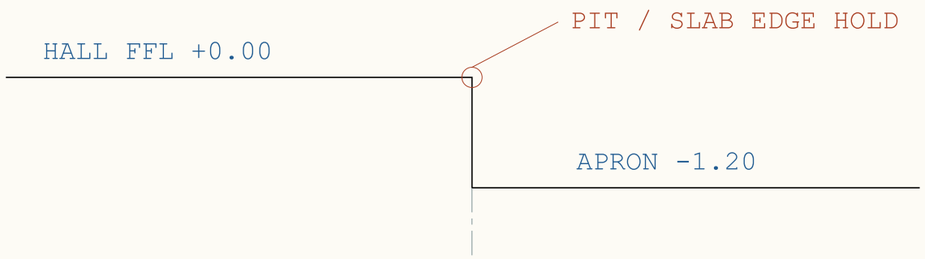
HALL-LOCAL COORDINATE REGISTER / M

SUPPORT	HALL X	HALL Y
F01 / 02	52.22	36.75 / 37.95
F03 / 04	54.18	36.75 / 37.95
F05 / 06	54.58	36.75 / 37.95
F07 / 08	55.82	36.75 / 37.95
F09 / 10	56.22	36.75 / 37.95
F11 / 12	58.18	36.75 / 37.95
F13	54.50	38.31
F14	55.90	38.31
F15	54.50	40.20
F16	55.90	40.20

TRANSFORM

Vector M120 centres + hall (52.00,36.65).
Axes parallel per P102. Vendor confirmation required.
Paired IDs are listed in lower-to-upper Y order.
Centre symbols indicate candidate supports only.

02 DOCK EDGE / RELEASE SECTION NTS



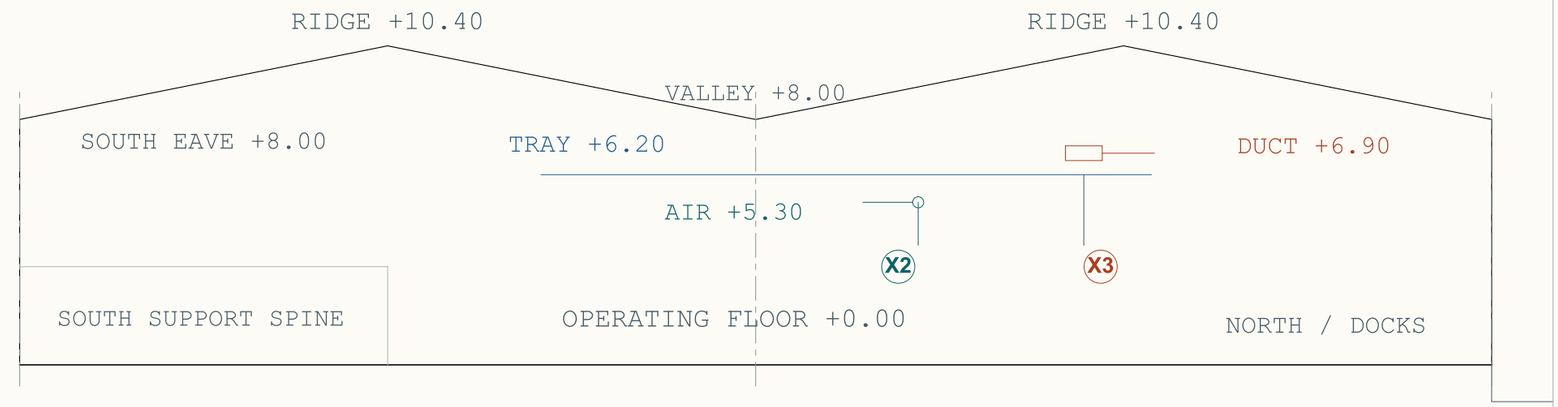
RELEASE HOLD / DO NOT DRILL

Obtain vendor reactions and approved anchor template.
Verify slab, joints, scanned obstructions and anchorage.
Confirm dock pit reinforcement, leveller and edge hardware.
Cross-check A301 / S102 and Vector M120 / M220.

Overhead service installation

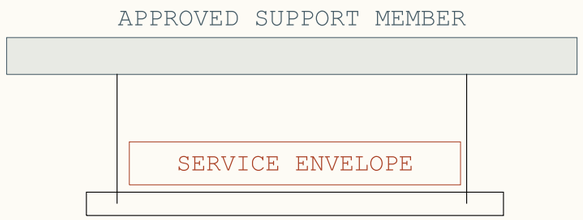
SECTION AT HALL X58 / NOMINAL SERVICE CENTRELINES / TYPICAL INTERFACES

01 SOUTH TO NORTH / X58



ROOF PROFILE PROJECTED FROM S201 / SERVICE SIZES NOT SET

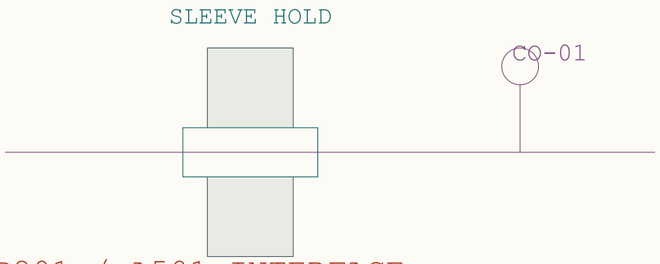
02 HANGER / SERVICE RESERVATION



INSTALLATION HOLD

Confirm hanger and attachment load.
Keep structure and access zones clear.

03 SANITARY PASSAGE / ACCESS



P201 / A501 INTERFACE

Check sleeve and access before set-out.
Approve firestop where barrier requires it.

CROSSING REGISTER / M101

ID	HALL XY m	SERVICES
X1	29, 29.3	Tray / process air
X2	58, 29.3	Tray / process air
X3	58, 34.7	Supply duct / tray

NOMINAL CENTRELINES

X1 and X2: tray +6.20 / air +5.30 m.
X3: duct +6.90 / tray +6.20 m.
Outside sizes and clear gaps held.

STRUCTURE COORDINATION

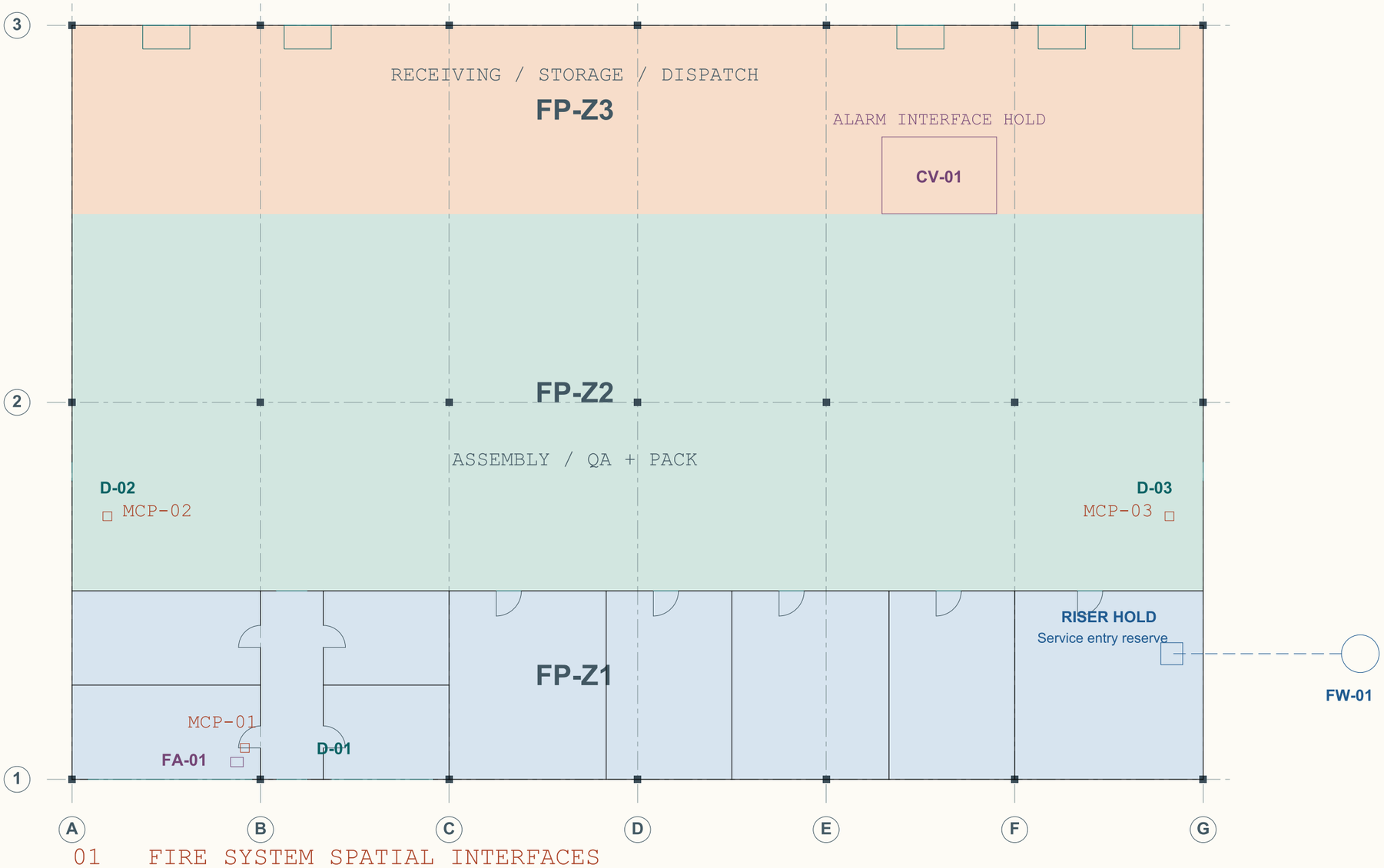
Check portal haunch and bracing zone.
Confirm loads, insert positions and service access with S201 / S202.

CHECK BEFORE RELEASE

Outside sizes, insulation and supports.
Sprinkler and luminaire obstructions.
Maintenance and lifting clearances.
Firestop system and access approval.

Fire protection installation interfaces

FIRE-WATER TIE-IN / SPATIAL ZONES / ALARM AND SHUTDOWN HANDOFF



01 FIRE SYSTEM SPATIAL INTERFACES

ZONE REGISTER

TAG	HALL BAND
FP-Z1	Y 0–12 m / staff
FP-Z2	Y 12–36 m / work
FP-Z3	Y 36–48 m / storage

WATER RESERVE

FW-01 site (112, 30) m.
C102 pad and F201 riser.
Confirm routing and capacity.

ALARM HANDOFF

MCP and waterflow to FA-01.
Review notify / HVAC / CV-01.
Validate each approved action.

SPATIAL CHECK

Verify rack heights and commodities.
Check services and obstructions.
Confirm authority and insurer basis.

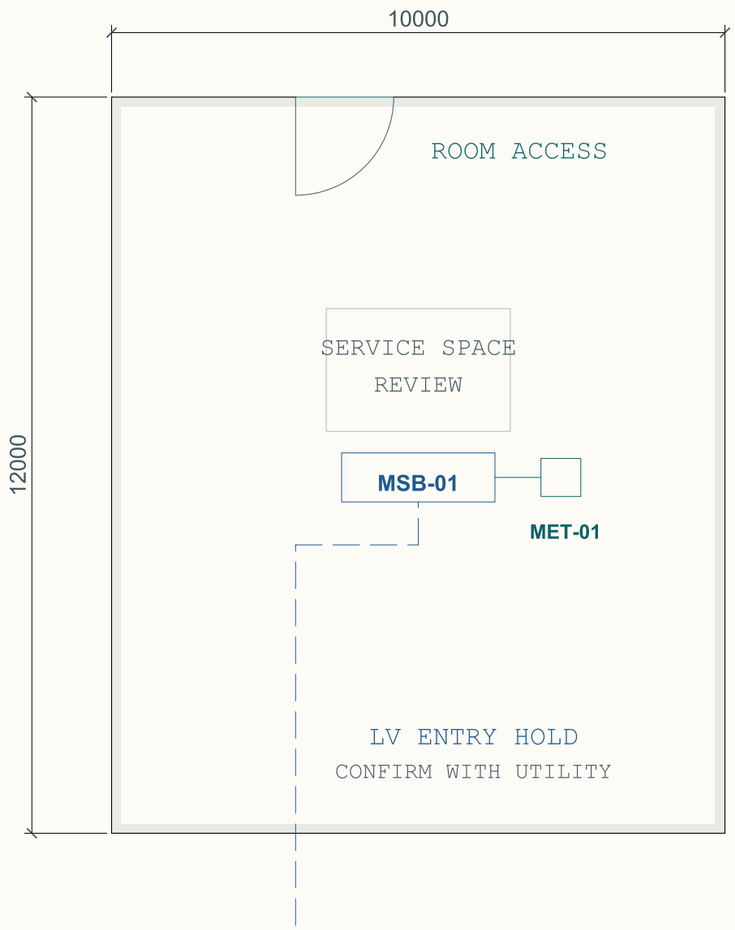
DRAWING PAIR

F101 egress / F201 functions.
Design and hydraulics held.

Electrical room and field interfaces

ROOM ACCESS / LV SERVICE ENTRY HOLD / FEEDER AND PROTECTIVE BONDING HANDOFF

01 ELECTRICAL ROOM / HALL X24-34



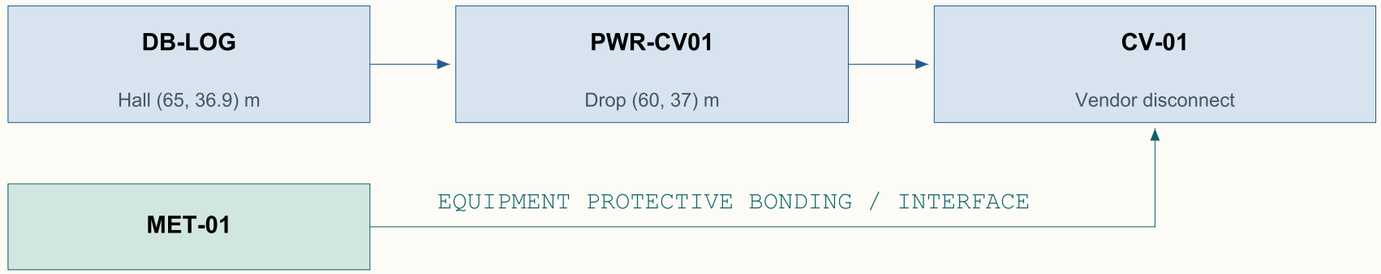
Equipment shapes indicate coordination only.

Maintain clear access from the support spine.

02 FEEDER HANDOFF / E601-E602

REF	DEVICE	FUNCTION	HOLD
Q0	MSB-01	Service and metering	Supply / fault data
Q1	DB-PROD	Assembly / QA / packing	Demand and route
Q2	DB-LOG	Docks / staging / CV-01	Disconnect and ratings
Q3	DB-ADM	Staff / communications	Circuit allocation
Q4	MCC-01	Air and HVAC plant	Motor and protection

03 CV-01 POWER PATH AND BONDING

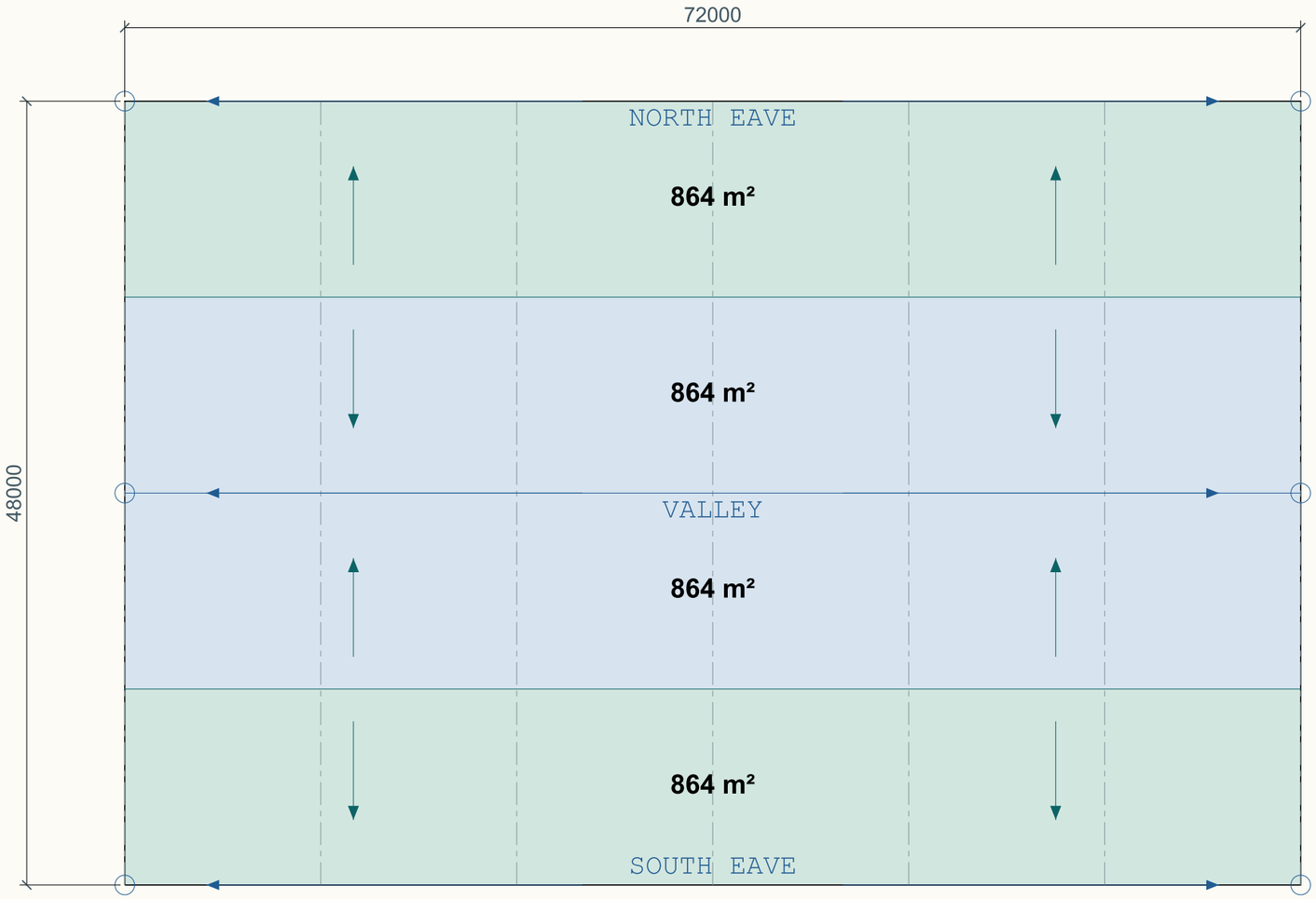


INSTALLATION / TEST HOLDS

Confirm utility source, available fault current and earthing system.
Verify panel access, containment and disconnect location.
Confirm protective continuity, insulation test and commissioning procedures.
Cable sizes, device ratings and termination points require approved vendor data.

Geometry and runoff study checks

REV 01 DIMENSION CHECK / PROJECTED ROOF CATCHMENTS / RAINFALL SENSITIVITY



01 PROJECTED ROOF AREA AND COLLECTION

Six outlet locations from S101. Split to the two ends is a trial assumption.

Stormwater boundaries and treatment references on C501.

Method reference: FHWA HEC-22, 4th ed. / Rational Method.

02 GEOMETRY CHECK

ITEM	FROM REV 01
Building plan	72 × 48 m / 3,456 m²
Column grid	7 × 3 / 21 points
Roof rise / half span	2.40 / 12.00 m
Nominal pitch	1:5 / 11.3°

03 TRIAL FLOW / L/s

COLLECTOR	AREA m²	75	100	150
S eave	864	18.0	24.0	36.0
Valley	1,728	36.0	48.0	72.0
N eave	864	18.0	24.0	36.0
Roof total	3,456	72.0	96.0	144.0

PORTFOLIO ASSUMPTIONS

Column headings are trial intensity in mm/h.
Q = C × i × area / 3600. Trial roof C = 1.0.
At 100 mm/h, equal end split is 12 / 24 / 12 L/s.

HOLD / HYDRAULIC DESIGN

Confirm storm, discharge, storage, sizing and overflow.